

Quantum and Classical Kernels under Distribution Shift: A Controlled Study of Kernel Geometry and Out-of-Distribution Robustness

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Abstract

Whether quantum kernels improve robustness under distribution shift is usually asked as a quantum-versus-classical question. We argue, and show empirically, that it is more productively asked as a question about kernel geometry. We present a controlled kernel-swap study in which the classifier, preprocessing, split construction, and family-internal model-selection logic are held fixed while only the kernel varies, instantiated on four security benchmarks spanning two modalities and three drift mechanisms (EMBER static malware; UNSW-NB15 and ToN-IoT network flows under synthetic covariate shift and natural regime drift), two classifier families (support vector classification and a Laplace-approximation Gaussian process classifier with calibrated uncertainty), and 72 principal settings with 15 repeated runs each; all quantum kernels are computed by exact statevector simulation, and no hardware claim is made. Against the customary linear and RBF baselines, fidelity-based quantum kernels improve out-of-distribution balanced accuracy nearly uniformly. This advantage, however, is a property of high-effective-rank, alignment-preserving geometry rather than of quantumness: once the classical family is enlarged with heavy-tailed kernels given the *same* length-scale tuning freedom, and configurations are selected without out-of-distribution labels (a deployment protocol on in-distribution data

and a cross-seed protocol), the advantage dissolves on EMBER and reverses on network flows, where the extended classical family wins the deployment comparison (hierarchical permutation test $p = 2 \times 10^{-4}$ under support vector classification). The quantum maps survive only against the restricted linear-plus-RBF baseline. A mechanistic analysis explains why: within each setting, out-of-distribution alignment is ordered by a single geometric axis—the effective rank of the training kernel—across both families alike (median Spearman $\rho \approx 0.93$ on EMBER, positive in 98% of units, and equally strong within the classical family alone), and the fidelity maps sit in the middle of this classical continuum. The mechanism survives label-permutation and cross-fitting controls that rule out circularity. What remains as a genuine quantum contribution is in-distribution separability, together with a short-length-scale classical kernel that reproduces the same robust geometry at a fraction of the cost. Beyond the specific verdict, the study contributes a reusable, fully reproducible protocol for attributing robustness effects to kernel geometry, and a design principle—prefer kernels whose label-aligned geometry survives shift—that spans classical and quantum families alike.

Keywords: quantum machine learning; quantum kernels; distribution shift; out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian processes; malware detection; network intrusion detection

1 Introduction

Quantum kernel methods are one of the clearest and most actively studied directions in quantum machine learning. By embedding classical data into quantum states and defining similarity through state overlaps, they provide a principled way to study how quantum-induced feature geometry affects supervised learning while keeping the downstream classifier fixed [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). At the same time, the recent QML literature has made an equally important point: empirical conclusions about quantum models can depend strongly on the data regime, the encoding, and the evaluation protocol [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#); [Gil-Fuster et al. \(2024\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#). For that reason, claims about the usefulness of quantum kernels are most informative when they are established under controlled, reproducible, and carefully matched comparisons rather than through heterogeneous pipelines or loosely aligned baselines.

This need for control becomes even more important under distribution shift. In many applied settings, the relevant question is not only whether a model separates classes well under in-distribution (ID) conditions, but whether its behavior remains reliable once the evaluation distribution changes. The broader distribution-shift and domain-generalization literature has repeatedly shown that apparent robustness gains can weaken, disappear, or become difficult to interpret when model-selection strategy, hyperparameter search, split construction, or evaluation standardization are not carefully controlled [Quiñonero-Candela et al. \(2008\)](#); [Zhou et al. \(2023\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). A convincing robustness study must therefore satisfy two requirements at once: it must isolate the factor of interest tightly enough to support attribution, and it must evaluate that factor broadly enough to determine whether the observed behavior reflects a meaningful regime-level effect rather than a narrow configuration artifact.

These issues are especially acute in security applications. Drift is not incidental to the problem; it is part of the problem definition itself. Static PE malware classification has been widely studied through benchmarks such as EMBER [Anderson and Roth \(2018\)](#), while more recent work has highlighted the gap between laboratory evaluation and performance on drifted malware, as well as the difficulty of adapting with limited post-drift data [Barbero et al. \(2022\)](#); [Li et al. \(2025\)](#). The emergence of updated resources such as EMBER2024 further reflects an active push toward more realistic, broader, and more deployment-relevant malware evaluation [Joyce et al. \(2025\)](#). Network intrusion detection faces the same structural challenge: traffic distributions move as services, infrastructure, and attack campaigns evolve, and public benchmarks such as UNSW-NB15 and ToN-IoT expose exactly this kind of regime change [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Taken together, this literature suggests that a useful security-ML study should not merely report accuracy on a convenient split; it

should examine how model choices behave when the evaluation explicitly targets shift, and it should do so across more than one data modality and drift mechanism.

Kernel methods are a natural setting in which to ask such questions cleanly. Because kernel-based support vector machines separate the choice of classifier from the choice of similarity function, they offer a principled way to study how representational geometry affects ID and OOD behavior while keeping the learning machinery fixed [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#). This perspective becomes particularly interesting in quantum machine learning, where quantum kernels are defined through implicit feature maps into quantum Hilbert spaces and estimated via state overlaps or related fidelity-based similarities [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). From this viewpoint, quantum kernels are not a vague alternative model class, but a concrete family of kernel constructions whose empirical value can—and should—be tested under the same standards applied to classical kernels.

At the same time, the quantum-kernel literature has matured to the point where optimistic demonstrations and critical scrutiny must now be read together. Foundational work established the feature-space interpretation of quantum kernels and motivated them as expressive alternatives to classical similarities [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#). Subsequent studies emphasized that performance in quantum machine learning can depend strongly on the data regime, the encoding, and the effective geometry induced by the model [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#). More recent work has sharpened this message further: understanding generalization in quantum machine learning requires care beyond classical intuition, and benchmarking conclusions can be highly sensitive to protocol choices [Gil-Fuster et al. \(2024\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#). This makes broad claims of superiority difficult to defend and reinforces the need for conservative, protocol-aware empirical studies.

The gap we address lies precisely at this intersection, and it has three parts. First, there remains limited evidence from controlled, variability-aware comparisons between classical and quantum kernel families in security-relevant settings where out-of-distribution robustness is part of the evaluation target: existing studies often differ simultaneously in model class, preprocessing, search space, or data protocol, making it difficult to attribute observed gains to the kernel itself. Second, when such comparisons are attempted, the classical reference set is typically narrow—linear and RBF kernels under a single classifier—so an apparent quantum advantage may reflect the weakness of the baselines rather than a property of the quantum family; probabilistic classifiers with principled uncertainty quantification, and classical kernels with richer spectral structure, are usually absent. Third, and most fundamentally, empirical wins and losses are rarely accompanied by an account of *why* a kernel family should behave better or worse under shift, leaving results as observations in search of an explanation. For a question as specific as whether quantum kernels help under distribution shift, these confounds and omissions are not peripheral; they are decisive.

In this paper, we study that question through a strictly controlled *kernel-swap* protocol, instantiated broadly enough to close all three gaps at once. Within each experimental setting we hold the classifier, the preprocessing pipeline, the split definitions, and the family-internal model-selection logic fixed, and vary only the kernel. The protocol is applied to four security benchmarks spanning two data modalities and three drift mechanisms: EMBER static PE malware under two synthetic covariate-shift constructions [Anderson and Roth \(2018\)](#), and UNSW-NB15 and ToN-IoT network-flow scenarios under both an analogous synthetic construction and *natural* regime drift, in which the out-of-distribution pool consists of genuinely drifted traffic [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Two classifier families consume the same precomputed kernels: support vector classification, and a binary Laplace-approximation Gaussian process classifier [Rasmussen and Williams \(2006\)](#) that adds

231 calibrated predictive probabilities—and hence principled uncertainty quantification
 232 under shift—without changing anything else in the pipeline. The classical family ranges
 233 from the customary linear and RBF baselines to polynomial, Laplacian, and Matérn
 234 kernels with train-only median-heuristic length scales; the quantum family comprises
 235 fidelity kernels induced by commonly used feature-map families. Crucially, both fam-
 236 ilies receive the same length-scale tuning freedom (a length-scale sweep for the RBF,
 237 Laplacian, and Matérn kernels, and an angle-scaling sweep for the quantum maps),
 238 so that neither side is compared at a fixed operating point the other is allowed to
 239 tune away from [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#), and configurations
 240 are then chosen by selection rules that do not consult the out-of-distribution labels.
 241 The main evaluation comprises 72 principal settings (18 on EMBER, 54 on network
 242 flows), each repeated across 15 runs, with paired Wilcoxon signed-rank tests and Holm
 243 correction on all family-level comparisons.

252 Beyond measuring who wins, we ask why. For every kernel, dimension, and setting
 253 we compute geometry descriptors of the induced Gram matrices—the effective rank of
 254 the training kernel [Roy and Vetterli \(2007\)](#), centered kernel-target alignment on the
 255 ID and OOD splits [Cristianini et al. \(2001\)](#); [Cortes et al. \(2012\)](#), kernel concentration
 256 statistics, and the geometric difference between classical and quantum kernels [Huang](#)
 257 [et al. \(2021\)](#)—and test which of these quantities predict out-of-distribution behavior
 258 within and across settings. This mechanistic layer converts the benchmark from a
 259 scoreboard into an instrument for attribution.

266 The resulting picture is sharper than a quantum-versus-classical verdict, and more
 267 useful. Against linear and RBF baselines, fidelity-based quantum kernels improve
 268 OOD balanced accuracy nearly uniformly across all four datasets. This advantage,
 269 however, does not survive a fair and honest comparison. When the classical family
 270 is enlarged with heavy-tailed kernels (polynomial, Laplacian, Matérn) granted the
 271 *same* length-scale tuning freedom the quantum maps receive through angle scaling,
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and configurations are selected without ever consulting OOD labels—by a deployment protocol that selects on in-distribution accuracy and a cross-seed protocol that selects on held-out shift seeds—the OOD advantage dissolves on EMBER and *reverses* on network flows, where the extended classical family wins the deployment comparison under a hierarchy-aware permutation test ($p = 2 \times 10^{-4}$ under SVC). The quantum family retains a significant OOD edge only against the restricted linear-plus-RBF baseline. A single geometric regularity explains the entire pattern: within a setting, OOD kernel-target alignment is ordered by the effective rank of the training kernel across both families alike (median within-setting Spearman $\rho \approx 0.93$ on EMBER, positive in 98% of units, and $\rho \approx 0.96$ within the classical family alone), and the fidelity maps sit squarely in the middle of this classical continuum rather than above it. The kernel that closes the gap is a Laplacian with a *shortened* length scale—exactly the classical kernel the mechanism predicts, because shortening the scale raises effective rank. What survives every control is narrower and more defensible: in-distribution separability favors the quantum family in every EMBER setting, and against the weak linear-plus-RBF baseline the quantum family wins throughout—but the property doing the work under shift is geometry, not quantumness.

Our contributions are fourfold. First, a controlled, reproducible kernel-swap protocol for robustness studies that extends across classifier families (including a probabilistic classifier with calibrated uncertainty), across a broad classical reference set with symmetric length-scale tuning, and across datasets, modalities, and drift mechanisms, evaluated under honest, no-ODD-label selection protocols and hierarchy-aware inference. Second, a variability-aware regime map of quantum-versus-classical kernel behavior under shift on 72 settings, whose central finding is a negative one obtained under fair conditions: once tuning freedom is symmetric and selection does not peek at the shifted test labels, the apparent OOD advantage of fidelity kernels over a

matched classical family dissolves on one modality and reverses on the other, surviving only against the customary weak baselines. Third, a mechanistic account—OOD alignment ordered by the effective rank of the training kernel, one axis common to both families—that predicts within-setting OOD behavior across every dataset we study, places the fidelity maps in the middle of a classical continuum, survives label-permutation and cross-fitting controls against circularity, and yields the checkable prediction (which we confirm) that a short-length-scale classical kernel reproduces the robust geometry. Fourth, practical guidance and a fully public artifact: for OOD-sensitive kernel selection, prefer kernels whose label-aligned geometry survives shift; a median-heuristic Laplacian with a shortened length scale delivers it classically, and fidelity-based quantum maps are one—costly—way to obtain the same geometry.

2 Related Work

2.1 Distribution shift, domain generalization, and evaluation rigor

The broader literature on distribution shift has established that predictive performance can degrade substantially when training and test data are drawn from different distributions, and that robust evaluation under shift requires more than reporting strong in-distribution performance [Quiñonero-Candela et al. \(2008\)](#); [Shimodaira \(2000\)](#); [Sugiyama et al. \(2007\)](#). Within machine learning, domain generalization (DG) has emerged as one of the main formal settings for studying out-of-distribution (OOD) performance without access to target-domain labels during training [Zhou et al. \(2023\)](#). This literature has also made clear that empirical conclusions about robustness are highly sensitive to evaluation design.

A particularly important development is the move toward benchmark-driven scrutiny of robustness claims: DomainBed showed that many reported domain-generalization improvements dissolve once datasets, architectures, hyperparameter

budgets, and model-selection criteria are held consistent [Gulrajani and Lopez-Paz \(2021\)](#); WILDS demonstrated that standard training leaves large OOD gaps under naturally occurring real-world shifts [Koh et al. \(2021\)](#); and Wild-Time established temporal shift as a distinct, practically relevant regime that existing methods struggle to close [Yao et al. \(2022\)](#). These works matter here not because security detection is a standard DG benchmark, but because they establish a methodological principle we adopt wholesale: when the goal is to compare robustness under shift, fairness of protocol is part of the contribution. Our comparison therefore fixes the classifier family, preprocessing, and split definitions, so that the role of the kernel can be interpreted with minimal confounding.

2.2 Security detection under drift and evolving evaluation benchmarks

In malware detection, the relevance of distribution shift is especially acute. Malware ecosystems evolve continuously as attackers modify payloads, packers, obfuscation strategies, delivery mechanisms, and development toolchains. Static detection models are therefore exposed to changing distributions in ways that are operationally consequential rather than merely academic. Public benchmarks have played an important role in making this problem more reproducible. EMBER remains one of the most widely used open datasets for static PE malware detection and helped standardize large-scale evaluation in this area [Anderson and Roth \(2018\)](#). More recent resources such as EMBER2024 push this effort further by broadening scope, increasing scale, and enabling more holistic evaluation across multiple malware-related tasks [Joyce et al. \(2025\)](#). Other datasets such as BODMAS and SOREL-20M likewise reflect the field’s effort to move toward larger and more realistic benchmark settings [Yang et al. \(2021\)](#); [Harang and Rudd \(2020\)](#).

At the same time, a growing body of work has shown that strong laboratory performance does not guarantee robustness to evolving malware. The Transcend line introduced conformal criteria to identify aging classifiers in deployment before their accuracy visibly degrades [Jordaney et al. \(2017\)](#), later refined by Barbero et al. [Barbero et al. \(2022\)](#); CADE detects and explains drifting samples without out-of-distribution labels [Yang et al. \(2021\)](#); and Li et al. studied adaptation to real drifted malware under limited post-drift supervision [Li et al. \(2025\)](#). On the methodological side, TESSER-ACT established that spatial and temporal experimental bias can manufacture illusory detector performance [Pendlebury et al. \(2019\)](#), and Arp et al. catalogued the recurring pitfalls of ML evaluations in computer security—sampling bias, label leakage, and inappropriate baselines among them [Arp et al. \(2022\)](#). Our protocol is, in large part, an application of those prescriptions to kernel comparison: fixed splits, leakage checks, temporal-style regime separation in the natural-drift construction, and baselines strengthened until they stop flattering the method under study. Taken together, these studies support a clear conclusion: in security detection, robustness under shift is not a niche desideratum but a primary evaluation target.

The same structural problem, with faster drift dynamics, defines network intrusion detection: benchmark corpora such as UNSW-NB15 and ToN-IoT capture traffic whose attack composition and background behavior change across collection regimes [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#), and adaptive NIDS pipelines are routinely retrained for exactly this reason. Using both modalities in one protocol lets us ask whether kernel-level robustness effects are properties of a dataset family or of the kernels.

Quantum machine learning has itself entered this application space. Quantum-enhanced classifiers and QSVMs have been applied to network intrusion detection since at least Gouveia and Correia [Gouveia and Correia \(2020\)](#), the area has been systematically reviewed [Nicesio et al. \(2023\)](#); [Nieto and Gil-Merino \(2026\)](#), and recent systems

evaluate quantum kernels directly on the benchmarks we use, reporting quantum advantages on UNSW-NB15 and ToN-IoT [Abreu et al. \(2024, 2025\)](#); a critical evaluation on PCA-based intrusion detection found classical methods to remain competitive under realistic constraints [Bellante et al. \(2025\)](#). A common trait of the optimistic studies is decisive for the present paper: they compare against a small set of classical baselines on i.i.d. splits—precisely the two evaluation choices our results show to inflate apparent quantum advantages. Our study can be read as the missing stress test for this literature: under matched baselines, symmetric tuning, and explicit shift, part of the advantage survives and part of it does not, and the geometry explains which.

Our work differs from this line in emphasis. Rather than proposing a new drift-adaptation mechanism, drift detector, or continual-update strategy, we ask a narrower but complementary question: under a tightly controlled protocol, does the choice of kernel family itself materially affect OOD robustness? This question is well motivated by the security literature, yet comparatively underexplored in a way that isolates kernels from the many other moving parts of the pipeline.

2.3 Kernel methods, quantum kernels, and careful benchmarking in QML

Kernel methods provide a natural framework for studying robustness under controlled model comparisons because they separate the learning algorithm from the similarity function used to compare inputs. In classical machine learning, support vector machines and related kernel methods have long provided a principled route for controlling model capacity while varying the geometry of the induced feature space [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#). This makes kernels especially attractive when the scientific question concerns representation and similarity rather than wholesale changes in classifier architecture.

Quantum kernel methods arise from the observation that data can be embedded into quantum states and compared through overlaps in a quantum feature space. Havlíček et al. popularized this perspective in the context of supervised classification with quantum-enhanced feature spaces [Havlíček et al. \(2019\)](#), while Schuld and Killoran provided a broader theoretical account linking quantum models to kernel methods through feature Hilbert spaces [Schuld and Killoran \(2019\)](#). Later work sharpened this connection by arguing that supervised quantum machine learning models can often be interpreted directly as kernel methods, making the kernel viewpoint central rather than incidental [Schuld \(2021\)](#). Additional work on data encoding further showed that representational behavior depends critically on how classical inputs are mapped into quantum states, which is directly relevant when comparing quantum feature maps as alternative kernel constructions [Schuld et al. \(2021\)](#). Two spectral results frame what such kernels can and cannot do: Kübler et al. showed that the inductive bias of quantum kernels is governed by the spectrum of the induced integral operator, with flat spectra implying poor in-distribution sample efficiency [Kübler et al. \(2021\)](#), and Thanasilp et al. established that broad families of fidelity kernels concentrate exponentially toward trivial values as systems grow, erasing trainability [Thanasilp et al. \(2024\)](#). Both results interact nontrivially with our findings and are taken up in the discussion.

However, the contemporary QML literature also makes clear that promising empirical behavior should not be read uncritically. Huang et al. showed that the role of data can be decisive in quantum machine learning, suggesting that observed performance gains are deeply regime-dependent rather than model-intrinsic [Huang et al. \(2021\)](#). Caro et al. studied generalization from few training examples [Caro et al. \(2022\)](#) and later proved out-of-distribution generalization guarantees for learning quantum dynamics [Caro et al. \(2023\)](#)—a setting whose inputs are quantum states rather than classical data, and thus complementary to the classical-input shift studied here—while

Gil-Fuster et al. argued that understanding generalization in QML requires rethinking familiar classical intuitions, since standard explanations may fail to account for observed behavior in quantum models [Gil-Fuster et al. \(2024\)](#). In parallel, broader perspective pieces have emphasized both the promise and the unresolved challenges of QML as a field [Cerezo et al. \(2022\)](#).

Most importantly for the present work, recent benchmarking studies have raised the standard for empirical claims in QML. Bowles et al. argued that experimental design choices can dominate conclusions and showed, in a broad simulated benchmark, that out-of-the-box classical models often remain strong baselines against quantum alternatives [Bowles et al. \(2024\)](#). Schnabel and Roth extended this benchmarking perspective specifically to quantum kernel methods, conducting a large-scale study across many design choices and showing that performance depends strongly on kernel type, encoding strategy, and hyperparameter configuration rather than on a simple classical-versus-quantum dichotomy [Schnabel and Roth \(2025\)](#). Related controlled evidence from adjacent QML settings points in the same methodological direction: recent work in hybrid quantum reinforcement learning shows that varying multiple pipeline components at once can obscure attribution, and highlights the importance of component-isolating, reproducible evaluation under fixed experimental conditions [Lazaro et al. \(2025\)](#). This line of work is especially relevant to our paper because it supports a conservative framing: the value of quantum kernels, if any, must be demonstrated under controlled and transparent comparisons, and should be expected to be regime-dependent.

Three further strands supply the analytical tools of this study. First, work on kernel bandwidth in QML has shown that rescaling the data encoding is as consequential for fidelity kernels as the length scale is for classical kernels, and can determine whether quantum models generalize at all [Shaydulin and Wild \(2022\)](#); [Canatar et al.](#)

(2023); Egginger et al. (2024); we operationalize this insight as a fairness requirement—symmetric bandwidth freedom for both families—rather than as a quantum-side tuning trick. Second, kernel-target alignment Cristianini et al. (2001); Cortes et al. (2012) and effective rank Roy and Vetterli (2007) are classical tools for characterizing Gram-matrix geometry, and the theory of spectral bias and task-model alignment shows that in-distribution generalization of kernel methods is governed precisely by the compatibility between the kernel’s spectrum and the target function Bordelon et al. (2020); Canatar et al. (2021); alignment has also served as a training objective for quantum embedding kernels Hubregtsen et al. (2022), a line our negative result on a-priori alignment-based selection speaks to directly. To our knowledge, the joint use of these descriptors as *predictors of OOD behavior under distribution shift*, across classical and quantum kernels alike, is new: we study the shift-facing counterpart of task-model alignment—whether the aligned spectrum survives once the distribution moves. Third, Gaussian process classification Rasmussen and Williams (2006) provides the standard route to calibrated uncertainty in kernel methods; including a Laplace-approximation GP classifier in the kernel-swap protocol lets us evaluate uncertainty quality under shift without changing anything else in the pipeline.

Finally, we note concurrent work. A 2026 preprint studies the robustness of quantum kernel methods under additive input corruption and proposes a spectral phase encoding to mitigate it Herrero Gómez et al. (2026). It shares our move away from clean-data peak performance, but differs in all three defining aspects: it perturbs inputs with synthetic noise rather than evaluating under covariate or natural regime shift, it proposes an encoding rather than auditing families under a fixed protocol, and it does not use Gram-matrix geometry as the explanatory variable. Large-scale quantum-kernel benchmarking without varied train–test distributions Bowles et al. (2024); Schnabel and Roth (2025) remains the closest genre; the shift axis and the mechanism are the present paper’s additions to it.

Our study is positioned at the intersection of these literatures. From the distribution-shift literature, we adopt the principle that robustness claims must rest on evaluation protocols that control major confounds. From the security literature, we take the practical motivation that temporal and distributional change are intrinsic to realistic deployment, in malware and network intrusion detection alike. From the QML literature, we take both the motivation for studying quantum kernels and the caution that benchmarking quality is central to interpretation. Relative to prior work, our contribution is not to claim universal superiority for quantum kernels, nor to propose a new adaptation method for drifted security data, but to provide a controlled, variability-aware, multi-dataset comparison of classical and quantum kernel families in which out-of-distribution robustness is the primary evaluation target—and to accompany the comparison with a mechanistic account of why its outcomes are what they are.

3 Methods

3.1 Problem setting

We study binary security classification—malware versus benign software, and attack versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution data-generating process and D_{OOD} an out-of-distribution process induced by a shift mechanism, which may be synthetically constructed or naturally occurring. Given a training set $T \sim D_{\text{ID}}$, we evaluate the learned classifier on both an ID test set S_{ID} and an OOD test set S_{OOD} . Our objective is not only to measure separability under the i.i.d. assumption, but to compare how kernel families behave when generalization is evaluated explicitly under shift.

The central methodological requirement of the study is attribution. Because the scientific question concerns the effect of the kernel, comparisons must avoid confounds from changes in classifier family, preprocessing, optimization protocol, or split design.

We therefore adopt a strictly controlled *kernel-swap* protocol in which all components of the pipeline are held fixed except the kernel function itself. This design follows the same evaluation logic that motivates rigorous OOD benchmarking in the broader distribution-shift literature: when the target of inference is robustness, protocol control is not an implementation detail but part of the contribution.

3.2 Datasets and split construction

The protocol is instantiated on four benchmark scenarios spanning two data modalities. The first is EMBER, the standard public benchmark for static PE malware detection [Anderson and Roth \(2018\)](#), which is widely used, supports reproducible feature-based experimentation, and has motivated later benchmark extensions such as EMBER2024 [Joyce et al. \(2025\)](#). The remaining three are network-flow intrusion scenarios: two derived from UNSW-NB15 (with DoS and Reconnaissance, respectively, as the drifting attack category) [Moustafa and Slay \(2015\)](#) and one from ToN-IoT (Scanning) [Alsaedi et al. \(2020\)](#). In all cases our goal is not to benchmark absolute detection performance on a standard split, but to construct controlled ID/OOD regimes that allow kernel-family comparisons under distribution shift.

For each experimental setting, we adopt a fixed three-way split design:

$$T, \quad S_{\text{ID}}, \quad S_{\text{OOD}},$$

where T is used for training, S_{ID} measures in-distribution performance, and S_{OOD} measures robustness under the induced shift. All models compared within a setting use exactly the same split definitions. This point is essential: a classical and a quantum model are never compared on different train/test partitions.

EMBER shift constructions.

On EMBER we operationalize shift through two controlled OOD-construction variants, denoted $m1$ and $m2$, both built on the histogram + byte-entropy feature blocks and both designed to preserve class balance across splits. Both are *label-conditional tail shifts*: the OOD pool is assembled from the extremes of a within-class score distribution, so the construction uses the labels to decide which samples are held out. We use this term rather than “covariate shift” deliberately. Because selection is conditioned on the class, we cannot assert that $P(Y | X)$ is preserved, which is what the covariate-shift assumption requires; class balance is preserved by construction, but that is a weaker property. The constructions are reproducible and interpretable stress tests of kernel geometry, not certified instances of a formal shift model, and none of our conclusions depend on their being so.

Variant $m1$: sparsity-score extremes within class.

For each sample, we compute a sparsity-related score on a predefined feature block. In the main configuration used in the experiments, this score corresponds to the number of non-zero entries within the selected block. The OOD set is formed by taking, within each class separately, the low-score and high-score extremes, i.e., the two tails of the within-class score distribution. The remaining samples form an ID pool from which the training and ID test sets are obtained by stratified sampling. Intuitively, this construction creates an OOD regime driven by atypical sparsity structure while keeping the label distribution under control.

Variant $m2$: distance-to-train extremes within class.

We first fix the final training set at an exact target size. Using only this training set, we compute class-conditional centroids and assign each non-training sample a score equal to its Euclidean distance to the centroid of its own class. To reduce noise and avoid leakage, this distance can be computed in a TruncatedSVD-projected space

783 fitted on the training set only. The OOD set is then defined, again within each class
784 separately, by the low- and high-distance extremes from the non-training pool, with
785 exact target OOD size and proportional class allocation. The ID test set is drawn from
786 the remainder. This produces a complementary shift regime in which OOD samples
787 are unusual relative to the final training geometry rather than to a sparsity statistic
788 alone.
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792 ***Network-flow scenarios and shift constructions.***

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794 Each network scenario is defined by a *reference* pool (traffic from the pre-drift regime)
795 and a *current* pool (traffic containing the drifted attack campaign), prepared with
796 a numeric-only feature policy that removes identifiers, label-derived columns, and
797 categorical protocol fields; UNSW-NB15 scenarios retain 39 flow features and ToN-IoT
798 89. We verified explicitly that no feature is label-derived: the maximum feature-label
799 correlation across scenarios is 0.62 (attained by well-known legitimate features such
800 as `sttl`), far from the near-perfect correlation that would signal leakage, and the
801 verification script is part of the public artifact. On these pools we instantiate two
802 shift mechanisms. The first, *m2-centroid*, is the direct analog of EMBER’s *m2*: within
803 the class-balanced reference pool, a provisional training pool is fixed, class-conditional
804 centroids are computed in a train-only MaxAbs+TruncatedSVD space, and the OOD
805 pool consists of the within-class low- and high-distance tails. The second, which we call
806 *attack-campaign regime shift*, applies no geometric selection: the training and ID pools
807 are drawn from the reference regime and the OOD pool from the class-balanced current
808 regime. The pools themselves are constructed—and we state the criterion precisely,
809 since it determines what the mechanism is being tested against. The reference pool
810 contains benign traffic plus every attack category *except* the target campaign; the
811 current pool contains benign traffic plus the target campaign. For UNSW-NB15 the
812 two pools additionally come from the two different official partitions of the testbed
813 capture. The OOD regime therefore combines an emerging attack campaign, absent
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from training, with a capture-partition change, and it is not a temporal split in the timestamp sense. Class priors are balanced separately in both pools, so prevalence shift is deliberately excluded and the measured effect is attributable to the change in traffic composition rather than to base rates. This is the most operationally realistic of our three mechanisms, and the least synthetic, but it is not free of construction; the full raw-to-pool pipeline is part of the public artifact.

Together, these constructions are not intended as universal models of real-world drift. Rather, they provide reproducible shift mechanisms of three distinct kinds—atypical-sparsity extremes, train-geometry extremes, and natural regime drift—that are sufficiently different to support regime-level conclusions while preserving comparability across kernel families. In every setting, all models are compared on exactly the same T , S_{ID} , and S_{OOD} .

3.3 Controlled kernel-swap protocol

Our core experimental principle is to isolate the kernel as the only variable of interest; Figure 1 summarizes the resulting pipeline. All compared models share the following components:

- **Classifier:** within each comparison, a single classifier consuming precomputed Gram matrices. Two classifier families are studied in parallel: Support Vector Classification with `kernel="precomputed"`, and a binary Gaussian process classifier under the Laplace approximation.
- **Preprocessing:** the same scaling and dimensionality-reduction pipeline for all classical and quantum models.
- **Split design:** identical T , S_{ID} , and S_{OOD} within each setting.
- **Selection logic:** the same family-internal model-selection principle applied consistently to classical and quantum candidates.

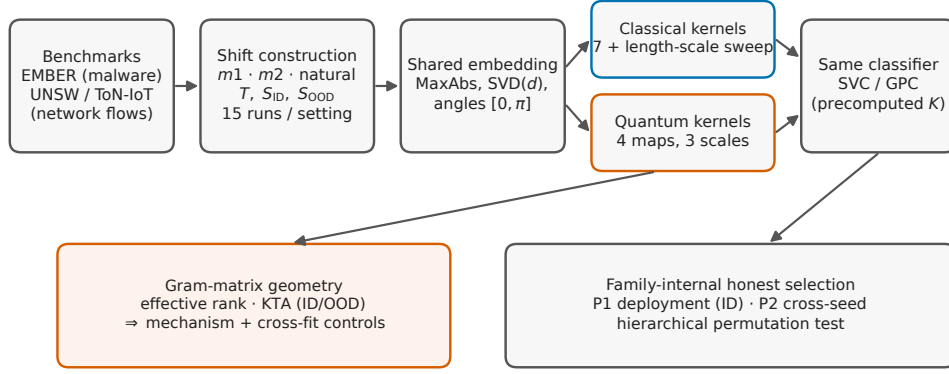


Fig. 1 The controlled kernel-swap protocol. Benchmark pools feed reproducible shift constructions; all kernels operate on a shared train-only embedding and are consumed by the same precomputed-kernel classifiers. Family-level conclusions are drawn by family-internal selection on run means with paired statistical testing, while the Gram matrices themselves feed the kernel-geometry analysis of Section 3.8.

This yields a comparison between kernel families rather than between heterogeneous model pipelines. In particular, it prevents observed OOD differences from being attributed to changes in classifier architecture, feature engineering, or split construction. The choice of precomputed-kernel classifiers is deliberate: it allows both classical and quantum similarities to be consumed by exactly the same decision rule, thereby making the comparison interpretable in kernel-method terms [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#).

The Gaussian process classifier implements the standard binary Laplace approximation with a logistic link (Algorithms 3.1 and 3.2 in [Rasmussen and Williams \(2006\)](#)), applied directly to the same unit-diagonal precomputed kernels as the SVC. Its role in the study is twofold. Methodologically, it tests whether the kernel-family conclusions are classifier-specific or transfer across learning machinery. Substantively, it produces predictive probabilities rather than scores, which allows robustness under shift to be examined through the lens of uncertainty quantification—log-loss, Brier score, expected calibration error, and predictive entropy on ID versus OOD data—rather than through accuracy alone. We stress that no post-hoc calibration procedure

(temperature scaling, isotonic regression, Platt scaling) is applied: the probabilities are those the Laplace approximation produces, and we *assess* their calibration rather than assert it. Kernel-swap comparisons are always performed within a classifier, never across classifiers.

Although the classical and quantum candidate sets are not identical in cardinality, fairness in this study is defined at the protocol level: both families are evaluated under the same classifier, preprocessing, splits, dimensionality sweep, and family-internal selection criterion, so that the effect attributed to the kernel is not confounded by changes elsewhere in the pipeline.

3.4 Preprocessing and feature representation

All models use the same preprocessing pipeline. Let $x \in \mathbb{R}^p$ denote the original feature vector of the dataset at hand. We apply the following steps:

1. **MaxAbs scaling.** We first apply `MaxAbsScaler` to normalize feature magnitudes while preserving sparsity structure.
2. **Dimensionality reduction.** We project the scaled features using `TruncatedSVD` to a reduced representation of dimension

$$d \in \{4, 6, 8, 10, 12\}.$$

3. **Angle mapping.** For the quantum pipeline, the reduced coordinates are linearly mapped to an angle range $[0, \pi]$ before being passed to the quantum feature maps.

This shared preprocessing serves two purposes. First, it keeps the comparison fair by ensuring that classical and quantum kernels operate on the same reduced representation. Second, it makes representation size itself a controlled factor of the study, enabling both aggregated setting-level comparisons and dimension-level analyses. The

dimensionality parameter d is therefore not a tuning artifact hidden in preprocessing,
but an explicit axis of the experimental grid.

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971 **3.5 Kernel families**

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973 ***Baseline classical kernels.***

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975 The customary reference set consists of two standard baselines:

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$$K_{\text{lin}}(x, x') = x^\top x', \quad (1)$$

980

981

$$K_{\text{rbf}}(x, x') = \exp(-\gamma \|x - x'\|_2^2), \quad (2)$$

982

983

984 with the RBF kernel evaluated using the standard `scale` convention for γ and the
985 linear kernel cosine-normalized to unit diagonal. These are the baselines against which
986 quantum kernels are most commonly compared, and they define our *classical-original*
987 family.

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Extended classical family.

Because an apparent quantum advantage over two baselines may say more about
the baselines than about the quantum family, we additionally evaluate a broader
classical-extended family: polynomial kernels of degree 2 and 3 (cosine-normalized),
the Laplacian kernel $K(x, x') = \exp(-\|x - x'\|_1/\ell_1)$, and Matérn kernels with
 $\nu \in \{3/2, 5/2\}$, where all length scales ℓ are set by the median heuristic—the median
pairwise distance on the training set, the standard choice in kernel two-sample test-
ing [Gretton et al. \(2012\)](#)—so that every hyperparameter is determined from training
data alone, without access to ID or OOD test information. These kernels were chosen
deliberately to span a range of spectral behaviors: the Laplacian and low- ν Matérn
kernels are less smooth than the RBF and induce Gram matrices with substantially
heavier eigenvalue tails, which, as the mechanistic analysis will show, is precisely the
property at stake under shift.

Bandwidth symmetry.

Kernel bandwidth is known to be a decisive hyperparameter for quantum kernels: rescaling the input encoding plays the same role for fidelity kernels that γ plays for the RBF kernel, and can qualitatively change generalization [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#). A comparison in which the classical family may tune its bandwidth while the quantum family is pinned to a fixed encoding scale (or vice versa) is therefore not protocol-fair. In the bandwidth-symmetric portion of the study, every kernel with a length scale is allowed to tune it: the RBF kernel is evaluated at $\gamma = f \cdot \gamma_{\text{scale}}$ and the Laplacian and Matérn kernels at length scales $\ell = f \cdot \ell_{\text{median}}$ for $f \in \{0.1, 0.3, 1, 3, 10\}$, while every quantum feature map is evaluated at angle scalings $\{0.5, 1, 2\}$ of the canonical $[0, \pi]$ encoding. This is stricter than tuning the RBF alone: the heavy-tailed classical kernels that most resemble the fidelity maps’ geometry are precisely the ones granted a length-scale sweep. Family-internal selection then operates over the enlarged candidate sets—115 classical against 60 quantum configurations—exactly as before.

Quantum kernels.

Quantum kernels are defined through an implicit feature map $\phi_Q(\cdot)$ induced by a parameterized quantum circuit, with similarity measured through fidelity-like overlaps [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). For two inputs x and x' , the corresponding kernel entry can be written abstractly as

$$K_Q(x, x') = |\langle \phi_Q(x), \phi_Q(x') \rangle|^2,$$

or equivalently through the overlap of states prepared by input-dependent quantum feature maps.

In the experiments, we evaluate multiple commonly used fidelity-based feature-map families chosen to represent distinct inductive biases: ZZ-type maps, PauliXZ-type

maps, and Z-map variants. Concretely, the candidate set includes two ZZ configurations with different repetition depths, one PauliXZ configuration, and one Z-map configuration. We emphasize that the study is not attempting to exhaust the quantum-kernel design space; rather, it compares a representative set of commonly used fidelity-based constructions under a tightly controlled protocol. This framing is consistent with recent work arguing that empirical conclusions in QML are often regime-dependent and highly sensitive to protocol choices [Schuld et al. \(2021\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#).

3.6 Model selection and repeated-run protocol

A major source of ambiguity in robustness studies is the selection criterion. Strong ID performance does not necessarily imply strong OOD performance, and a criterion that consults the shifted test labels is not deployable. We therefore apply family-internal selection at three levels of stringency, identically to both families.

P3, Best-by-OOD (oracle).

Within each family we select the configuration with the highest mean OOD balanced accuracy and compare the selected representatives. Because it peeks at the target split, this is an oracle upper bound rather than a deployable rule; it is the comparison that dominates the optimistic literature, and we report it as such.

P1, deployment (honest).

Within each run and family we select the configuration with the highest *in-distribution* balanced accuracy and report its OOD accuracy. No OOD labels enter the selection, so P1 is a rule a practitioner could actually deploy; it is our primary comparison.

P2, cross-seed (honest).

For each setting we perform leave-one-seed-out selection over the five q-split seeds: pick the configuration with the best mean OOD accuracy on four seeds and evaluate

it on the held-out seed. Shift labels inform the choice only from splits disjoint from the evaluation split, answering whether a robust kernel can be chosen from prior examples of the same shift.

Best-by-ID.

Within each family we also select by ID balanced accuracy and compare, as a separability-oriented view reported separately to avoid conflating ID accuracy with OOD robustness.

This separation is methodological, not cosmetic. It acts as a guardrail against overinterpreting ID gains as robustness gains, and against reading an oracle comparison as a deployable one; it disentangles which kernel family yields the strongest in-distribution class separation from which yields the strongest, honestly selectable out-of-distribution generalization.

Selection is only one source of ambiguity in robustness studies; a second is result stability. Each configuration is therefore repeated across 15 runs, generated by combining five q-split seeds with three model seeds under fixed setting definitions (shift variant, master seed, and size regime). Within a setting, both classical and quantum candidates are evaluated on identical split definitions and summarized by mean $\pm$ standard deviation over the repeated runs, so that every reported comparison is variability-aware while preserving strict comparability between kernel families.

Statistical testing.

Family-level comparisons are tested with paired two-sided Wilcoxon signed-rank tests over the per-setting deltas (the settings are the paired units), separately for each classifier, scenario, and selection view, with Holm correction. Because settings within a dataset are correlated, cross-dataset claims are based instead on a hierarchy-aware sign-flip permutation test whose statistic weights each dataset equally, reported with

leave-one-dataset-out ranges and bootstrap confidence intervals rather than by pooling correlated settings as independent replicates. We retain the per-setting effect-size proxy defined below as a descriptor of magnitude relative to run-to-run noise, not as an inferential criterion.

3.7 Evaluation metrics

Our primary metric is balanced accuracy, reported separately on ID and OOD test sets:

$$\text{BAcc}_{\text{ID}}, \quad \text{BAcc}_{\text{OOD}}.$$

Balanced accuracy is preferred because it remains informative under moderate class imbalance and is straightforward to interpret in both settings.

The ID-to-OOD degradation

$$\text{drop} = \text{BAcc}_{\text{ID}} - \text{BAcc}_{\text{OOD}}$$

is recorded for every configuration and available in the artifact; because it conflates OOD behavior with the corresponding ID baseline, we treat OOD balanced accuracy as the primary inferential target and use drop only as a secondary descriptor.

To compare families after selection, we compute family deltas:

$$\Delta_{\text{OOD}} = \text{BAcc}_{\text{OOD}}^{(Q)} - \text{BAcc}_{\text{OOD}}^{(C)}, \tag{3}$$

$$\Delta_{\text{ID}} = \text{BAcc}_{\text{ID}}^{(Q)} - \text{BAcc}_{\text{ID}}^{(C)}. \tag{4}$$

Positive values indicate an advantage for the selected quantum model relative to the selected classical model.

Because average gains can be misleading when variability is large, we also report an effect-size-to-variability proxy:

$$\text{effect} = \frac{\mu_Q - \mu_C}{\sqrt{\sigma_Q^2 + \sigma_C^2}},$$

where μ_Q, μ_C and σ_Q, σ_C are the repeated-run means and standard deviations for the metric of interest. Values greater than 1 indicate that the observed mean gain exceeds the combined variability scale, which provides a simple and interpretable robustness check on whether an apparent gain is large relative to run-to-run noise.

Probabilistic metrics.

For the Gaussian process classifier we additionally report, on both ID and OOD splits, the negative log-likelihood (log-loss), the Brier score, the expected calibration error [Guo et al. \(2017\)](#) (ECE, 15 equal-width bins), and the mean predictive entropy. These quantities characterize how the quality and honesty of predictive uncertainty change under shift: a well-behaved probabilistic model should become *more* uncertain on drifted inputs (rising entropy) while its calibration degrades as little as possible (bounded ECE increase), the desideratum established by the benchmark of Ovadia et al. [Ovadia et al. \(2019\)](#).

3.8 Kernel-geometry descriptors

To move from measuring outcomes to explaining them, we compute geometry descriptors of the Gram matrices that every kernel induces in every setting and dimension. All descriptors are computed from the very kernel matrices consumed by the classifiers (unit diagonal by construction), so the analysis describes exactly the objects the models see.

1243 ***Effective rank.***

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1245 For the training Gram matrix K_{tr} with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ normalized

1246

1247 to $p_i = \lambda_i / \sum_j \lambda_j$, the effective rank is $\exp(-\sum_i p_i \log p_i)$ Roy and Vetterli (2007).

1248 It measures how many directions of feature space the kernel actually uses: a cosine-

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1250 normalized linear kernel on low-dimensional embeddings is near rank one, while fidelity

1251

1252 kernels can spread mass over many directions.

1253

1254 ***Centered kernel-target alignment.***

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1256 For a kernel block K on a labeled split with labels $y \in \{-1, +1\}^n$, the centered kernel-

1257

1258 target alignment (KTA) is $\langle K_c, yy^\top \rangle_F / (n \|K_c\|_F)$, where K_c is the doubly centered

1259

1260 kernel Cristianini et al. (2001); Cortes et al. (2012). We evaluate KTA on the training,

1261 ID, and OOD splits separately; the ID-to-OOD change in alignment measures whether

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1263 the label structure remains visible to the kernel’s geometry after shift—the quantity

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1265 we will call *alignment survival*.

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1267 ***Concentration and cross-split similarity.***

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1269 We record the mean and standard deviation of off-diagonal kernel entries within the

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1271 training block and of the cross blocks between each test split and the training set,

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1273 together with the ratio of mean OOD-to-train and ID-to-train similarities, which mea-

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1275 sures how novel the OOD inputs look under each kernel’s similarity notion; these

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1277 cross-split means are the elementary statistics underlying kernel two-sample tests such

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1279 as MMD Gretton et al. (2012), used here descriptively rather than as hypothesis tests.

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1280 ***Geometric difference.***

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1282 Between each classical training kernel K_C and each quantum training kernel K_Q

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1284 we compute the regularized geometric difference $g(K_C \rightarrow K_Q) = \|\sqrt{K_Q}(K_C +$

1285

1286 $\lambda n I)^{-1} \sqrt{K_Q}\|_\infty^{1/2}$ of Huang et al. Huang et al. (2021), over a sweep of relative regu-

1287

1288 larizers λ . Large values indicate that the quantum geometry can deviate substantially

from what the classical kernel expresses; the quantity bounds the *potential* for a performance separation but not its sign.

Mechanistic analyses.

These descriptors feed two complementary analyses. First, a diagnostic analysis at the setting level asks which descriptors of the Best-by-OOD-selected kernels separate classical-favorable from quantum-favorable settings (Mann–Whitney AUC over settings) and correlate with the observed OOD delta. Second, a within-setting analysis asks whether geometry predicts performance across *all* kernel–dimension cells of a setting: for every setting, classifier, and q-split seed we compute the Spearman rank correlation between a descriptor (effective rank, OOD alignment) and OOD balanced accuracy over the 30–55 evaluated cells, and then examine the distribution of these correlations across settings and datasets. A descriptor that yields consistently positive within-setting correlations across datasets, drift mechanisms, and classifiers constitutes a dataset-general regularity rather than a benchmark artifact.

3.9 Experimental grid and reporting

The EMBER evaluation grid comprises 18 principal settings formed by the Cartesian product of:

- two shift variants ($m1, m2$),
- three master seeds,
- and three training-size regimes.

The three size regimes are:

$$(1000, 500, 500), \quad (2000, 1000, 1000), \quad (4000, 1800, 1800),$$

corresponding to $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$.

Table 1 Compact summary of the controlled experimental protocol. The repeated-run design is generated by combining five split seeds with three model seeds, yielding 15 runs per evaluated configuration under fixed setting definitions.

Component	Specification
Datasets	EMBER; UNSW-NB15 (DoS, Reconnaissance); ToN-IoT (Scanning)
Shift constructions	EMBER: $m1, m2$; network: m2-centroid, natural drift
Master seeds	42, 123, 999
Split seeds	42, 123, 999, 7, 2024
Model seeds	42, 123, 999
Runs per configuration	$5 \times 3 = 15$
Size regimes ($n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}}$)	(1000, 500, 500), (2000, 1000, 1000), (4000, 1800, 1800)
Dimensions	$d \in \{4, 6, 8, 10, 12\}$
Classifiers	SVC (precomputed); Laplace-GP classifier (precomputed)
Shared preprocessing	MaxAbsScaler + TruncatedSVD + angle mapping
Classical-original family	Linear, RBF(scale)
Classical-extended family	+ poly(2,3), Laplacian, Matérn(3/2, 5/2), median heuristic
Quantum family	ZZ (1 rep), ZZ (2 reps), PauliXZ (1 rep), Z-map
Bandwidth sweep	RBF γ factors $\{0.1, 0.3, 1, 3, 10\}$; angle scales $\{0.5, 1, 2\}$
Selection	honest P1 (ID), P2 (cross-seed); oracle P3 (OOD); Best-by-ID
Statistical testing	Paired Wilcoxon (per scenario) + Holm; hierarchical permutation test
Principal settings	EMBER $2 \times 3 \times 3 = 18$; network $3 \times 2 \times 3 \times 3 = 54$; total 72

The network-flow grid applies the same structure to the three scenarios (UNSW-DoS, UNSW-Reconnaissance, ToN-IoT-Scanning), each under two shift mechanisms (m2-centroid and natural drift), three master seeds, and the same three size regimes, yielding 54 principal settings. Together the study comprises 72 principal settings, each evaluated across the dimension sweep and repeated over 15 runs, for both classifiers. The bandwidth-symmetric sweep enlarges the kernel candidate sets on the full EMBER grid and on the natural-drift network settings.

For clarity and reproducibility, Table 1 summarizes the main experimental protocol used throughout the paper, including the shift constructions, controlled seed structure, size regimes, dimensionality sweep, kernel families, classifiers, and repeated-run design.

Within each setting, we evaluate both kernel families across the shared preprocessing grid and the dimension sweep $d \in \{4, 6, 8, 10, 12\}$. This produces both a setting-level comparison, used for the honest-selection and oracle analyses, and a broader dimension-level view used to characterize how representation size interacts with kernel-family behavior.

For reproducibility, split definitions are fixed within each setting, random seeds are controlled, and all key quantities are reported as mean $\pm$ standard deviation across repeated runs. Throughout the paper, we deliberately avoid claims of universal dominance or complexity-theoretic quantum advantage. The object of study is narrower and more empirical: under a fixed classifier and a fixed evaluation protocol, does changing only the kernel alter robustness under distribution shift in security-relevant tasks—and which property of the kernel is responsible?

4 Results

4.1 Overview

We report results under the controlled kernel-swap protocol described in Section 3. All comparisons share the same classifier, preprocessing pipeline, and ID/OOD split definitions within a setting; the only varying component is the kernel, and every number below is a mean over 15 repeated runs. The argument proceeds in three movements. First, against the customary linear+RBF baselines the fidelity kernels show a near-uniform OOD advantage (Section 4.2)—the result that dominates the optimistic literature. Second, we subject that advantage to a fair test: we enlarge the classical family with heavy-tailed kernels given the same length-scale tuning freedom, and we select configurations without ever consulting OOD labels, using a deployment protocol (P1) and a cross-seed protocol (P2) in place of the test-peeking oracle (P3). Under this fair test the advantage dissolves on EMBER and reverses on network flows (Section 4.3). Third, a single geometric mechanism—OOD alignment ordered by the

effective rank of the training kernel, common to both families and robust to circularity controls—explains the entire pattern and predicts the classical kernel that closes the gap (Section 4.4). Section 4.5 then examines predictive uncertainty under shift.

4.2 EMBER versus the customary baselines: a near-uniform quantum advantage

We begin with the comparison that dominates the literature: fidelity-based quantum kernels against linear and RBF baselines. For each setting and classifier, family-internal Best-by-ODD selection picks the configuration with the highest mean OOD balanced accuracy on the run means, and the selected representatives are compared directly. Table 2 (upper panel, first and third rows) reports the outcome.

The advantage is close to uniform. Under SVC the selected quantum kernel improves OOD balanced accuracy in 17 of 18 settings (mean $\Delta_{\text{OOD}} = +0.037$, Holm-adjusted Wilcoxon $p = 7.6 \times 10^{-5}$), with the gain exceeding the combined run-to-run variability scale in 12 settings; under the GP classifier the corresponding figures are 15 of 18, $+0.028$, $p = 2.1 \times 10^{-4}$, and 11 settings. The few classical-favorable settings are marginal: all occur at the largest training size under the *m1* construction, with $|\Delta_{\text{OOD}}| \leq 0.010$ and effect sizes below one, i.e., within run-to-run noise. The dimension-level view confirms that this is not a best-of-grid artifact: comparing the best quantum against the best classical candidate separately at every setting–dimension cell, the quantum family is ahead in 88 of 90 cells under SVC and 85 of 90 under the GP classifier (mean $\Delta_{\text{OOD}} \approx +0.05$ in both cases).

Under Best-by-ID selection (lower panel), the quantum family improves ID balanced accuracy in all 18 settings under both classifiers, with mean gains of $+0.078$ (SVC) and $+0.065$ (GPC) and every setting exceeding the variability scale. We keep the two views separate throughout: strong ID separability is not evidence of robustness, and the next section shows that the two properties respond very differently

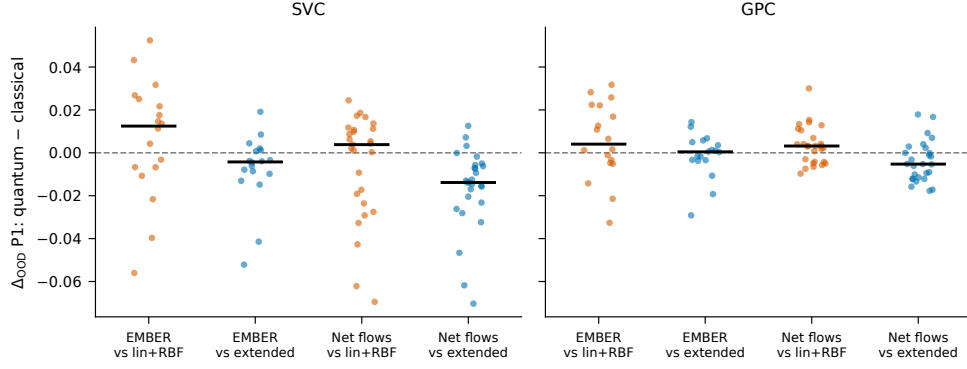


Fig. 2 The honest-selection reversal. Per-setting deployment-protocol (P1) deltas in OOD balanced accuracy (quantum minus best classical of the stated reference, selected without OOD labels; mean over 15 runs), for SVC (left) and the GP classifier (right); black bars mark medians. Against the customary linear+RBF baselines the quantum family stays ahead on both modalities; against the extended classical family—granted the same length-scale tuning freedom—the median delta falls to zero on EMBER and below zero on network flows. The apparent quantum OOD advantage is a property of the weak baseline, not of the quantum family.

once the classical baseline and the selection rule are made fair. The numbers in this subsection use the oracle Best-by-OOD selection (protocol P3), which peeks at the shifted test labels; we retain it as an upper bound and turn to honest, no-OOD-label selection next. Figure 2 previews the outcome: the per-setting deployment-protocol deltas against linear+RBF (first and third columns) stay above zero, but against the geometry-matched extended family (second and fourth columns) they sit at or below it, on both modalities.

4.3 A fair test dissolves the advantage on EMBER and reverses it on network flows

The comparison of Section 4.2 is unfair in two ways that both favor the quantum family, and correcting them changes the verdict. First, the classical reference set was pinned to linear and RBF, the two lowest-rank geometries in our study; a fair test must include classical kernels able to reach the same geometry, and must grant them

Table 2 EMBER oracle (Best-by-OOD / Best-by-ID) results at fixed classical bandwidth (18 settings, 15 runs each). Selection here peeks at the target split (protocol P3) and the classical family is untuned; both choices favor the quantum family, and Table 3 reports the honest, symmetrically tuned comparison. Within each classifier (GPC = Laplace Gaussian process classifier), family-internal selection compares the best quantum candidate against the best candidate of the stated classical reference set. p -values are two-sided paired Wilcoxon over settings, Holm-corrected within classifier; the last column counts settings whose quantum-favorable gain exceeds the combined run-to-run variability scale.

Selection	Classifier	Classical ref.	Q wins	Mean Δ	Median Δ	p (Holm)	Eff. > 1
Best-by-OOD	SVC	linear+RBF	17/18	+0.0374	+0.0426	7.6×10^{-5}	12
	SVC	extended	15/18	+0.0044	+0.0064	0.03	1
	GPC	linear+RBF	15/18	+0.0283	+0.0339	2.1×10^{-4}	11
	GPC	extended	11/18	+0.0023	+0.0029	0.26	0
Best-by-ID	SVC	linear+RBF	18/18	+0.0775	+0.0794	3.1×10^{-5}	18
	SVC	extended	18/18	+0.0149	+0.0152	3.1×10^{-5}	11
	GPC	linear+RBF	18/18	+0.0649	+0.0664	3.1×10^{-5}	18
	GPC	extended	18/18	+0.0192	+0.0166	3.1×10^{-5}	11

the same length-scale tuning freedom the quantum maps receive through angle scaling. Second, Best-by-OOD selection (protocol P3) chooses each configuration by its balanced accuracy on the very OOD split it then reports—an oracle that peeks at the shifted test labels. A deployable comparison must select without them.

We therefore adopt two honest selection protocols alongside the oracle. *P1* (*deployment*): within each run and family, select the configuration with the best in-distribution balanced accuracy and report its OOD accuracy; no OOD labels enter the choice. *P2* (*cross-seed*): for each setting, hold out one q-split seed, select the configuration with the best mean OOD accuracy on the remaining seeds, and evaluate on the held-out seed—using shift labels only from splits disjoint from the evaluation split. P3 is retained as an oracle upper bound. Under the enlarged, symmetrically tuned candidate sets the classical family offers 115 configurations against the quantum family’s 60, so a residual quantum win cannot be attributed to a larger search. Family-level deltas are aggregated by a sign-flip permutation test on the statistic T that weights datasets equally, with leave-one-dataset-out (LODO) ranges, rather than pooling correlated settings as independent replicates.

Table 3 EMBER honest-selection results (18 settings, 15 runs each). Hierarchy-aware mean delta T in OOD balanced accuracy (quantum minus best classical of the stated reference) with a sign-flip permutation p -value. P1 selects on in-distribution accuracy, P2 on held-out shift seeds, and P3 is the test-peeking oracle. Against the extended classical family the quantum advantage is absent under the honest protocols.

Classical ref.	Protocol	Classifier	Mean Δ	perm p
linear+RBF	P1 deployment	SVC	+0.0065	0.33
	P1 deployment	GPC	+0.0054	0.22
	P2 cross-seed	SVC	+0.0151	1.2×10^{-3}
	P2 cross-seed	GPC	+0.0116	5.0×10^{-3}
	P3 oracle	SVC	+0.0163	1.0×10^{-4}
	P3 oracle	GPC	+0.0134	3.0×10^{-4}
extended classical	P1 deployment	SVC	−0.0074	0.07
	P1 deployment	GPC	−0.0012	0.63
	P2 cross-seed	SVC	−0.0038	0.14
	P2 cross-seed	GPC	−0.0004	0.89
	P3 oracle	SVC	−0.0030	9.0×10^{-4}
	P3 oracle	GPC	+0.0008	0.58

EMBER.

Table 3 reports the hierarchy-aware deltas. Against linear+RBF the quantum family still wins under the label-using protocols (P2/P3, $p \leq 5 \times 10^{-3}$). Against the extended classical family, the advantage is gone: under the deployment protocol P1 the mean delta is -0.007 (SVC) and -0.001 (GPC), neither significant, and the oracle itself turns slightly negative under SVC (-0.003 , $p = 9 \times 10^{-4}$). The per-scenario picture is flat—on the sparsity construction $m1$ the P1 delta is negative (-0.013 , SVC), on the centroid construction $m2$ it is at best marginal ($+0.004$, GPC/P2, $p = 0.07$)—so the fixed-bandwidth “neutralization” by a single Laplacian reported in earlier accounts was in reality the quantum family already at parity once the classical side could tune its scale. The kernel that closes the gap is a Laplacian with a *shortened* length scale, selected in 21 of 33 classical Best-by-OOD choices; Section 4.4 shows this is exactly what the mechanism predicts.

Table 4 Network-flow honest-selection results (54 settings across UNSW-NB15 DoS/Reconnaissance and ToN-IoT Scanning, two shift mechanisms, 15 runs each). Hierarchy-aware mean delta T (quantum minus best classical), permutation p , and leave-one-dataset-out range. Against the extended classical family under the deployment protocol the delta is negative—the classical family wins.

Classical ref.	Protocol	Classifier	Mean Δ	perm p	LODO
linear+RBF	P1 deployment	SVC	-0.0062	0.20	[-0.0148, +0.0087]
	P1 deployment	GPC	+0.0035	0.05	[+0.0003, +0.0063]
	P2 cross-seed	SVC	+0.0083	1.7×10^{-3}	[+0.0027, +0.0136]
	P2 cross-seed	GPC	+0.0049	0.04	[+0.0036, +0.0059]
	P3 oracle	SVC	+0.0093	1.0×10^{-4}	[+0.0039, +0.0138]
	P3 oracle	GPC	+0.0080	1.0×10^{-4}	[+0.0066, +0.0099]
extended classical	P1 deployment	SVC	-0.0165	2.0×10^{-4}	[-0.0229, -0.0069]
	P1 deployment	GPC	-0.0040	0.04	[-0.0076, -0.0020]
	P2 cross-seed	SVC	-0.0039	0.05	[-0.0047, -0.0022]
	P2 cross-seed	GPC	-0.0046	0.04	[-0.0067, -0.0023]
	P3 oracle	SVC	-0.0026	0.01	[-0.0036, -0.0020]
	P3 oracle	GPC	-0.0007	0.52	[-0.0013, +0.0005]

Network flows.

Here the correction does more than dissolve the advantage; it reverses it (Table 4). The oracle P3 against linear+RBF still shows a large, uniform quantum win—the result reported in the optimistic literature—and against linear+RBF the quantum family retains a smaller edge under P2. But against the extended classical family under the deployment protocol P1, the extended family wins: the hierarchy-aware delta is -0.017 under SVC (permutation $p = 2 \times 10^{-4}$, LODO [-0.023, -0.007]) and -0.004 under GPC ($p = 0.04$), with every leave-one-dataset-out value negative. The reversal is significant in the scenarios where the classical kernel can most exploit a shortened scale—ToN-IoT Scanning (-0.036 SVC, -0.007 GPC, both $p \leq 4 \times 10^{-3}$) and UNSW-NB15 DoS under SVC (-0.010, $p = 8 \times 10^{-3}$)—and null elsewhere, with no scenario favoring the quantum family. The earlier finding that the quantum advantage “survives the extended family” on network flows was an artefact of the oracle selection and the fixed classical bandwidth; neither survives a fair test.

The verdict is an artefact of the comparison, not the physics.

Read together, Sections 4.2–4.3 show that the sign of the quantum-versus-classical OOD verdict is controlled by two protocol choices—the strength of the classical baseline and whether selection peeks at the shifted labels—and not by the quantum family itself. Against weak baselines and a test-peeking oracle the advantage looks large and uniform; against a geometry-matched family and an honest selection rule it is absent or reversed. The one comparison that survives every correction is in-distribution separability (Section 4.2), a statement about expressivity rather than robustness.

4.4 One geometric mechanism explains the pattern

The results so far pose a puzzle: why do quantum kernels beat linear and RBF baselines so uniformly, and why does a geometry-matched classical family—above all a short-length-scale Laplacian—erase and then reverse that advantage under a fair test? A single geometric quantity answers both. The effective rank of the training kernel orders out-of-distribution alignment identically across classical and quantum kernels, and the fidelity maps sit in the middle of the resulting continuum rather than above it.

A cross-dataset regularity.

Within every setting we correlate, across all evaluated kernel–dimension cells, each geometry descriptor with OOD balanced accuracy (Table 5 and Figure 3). The alignment-survival quantity—centered kernel-target alignment evaluated on the OOD split—predicts OOD accuracy within settings essentially everywhere: the within-setting Spearman correlation is positive in 89–99% of the 360 setting-seed units, with median ρ between 0.42 and 0.74, across all four datasets, both classifiers, and all three drift mechanisms. Effective rank of the training kernel—a train-only quantity—is an equally strong predictor on EMBER and ToN-IoT (median $\rho \approx 0.75$ –0.84), moderate on UNSW-DoS, and weak (though still predominantly positive) on UNSW-Reconnaissance. The natural reading is that high effective rank is the structural

Table 5 Within-setting Spearman correlation between kernel-geometry descriptors and OOD balanced accuracy, summarized per dataset and classifier over all setting-seed units (five q-split seeds per setting; 30–55 kernel–dimension cells per unit). “> 0” is the fraction of units with a positive correlation.

Dataset	Classifier	Units	Effective rank		KTA on OOD	
			median ρ	> 0	median ρ	> 0
EMBER	GP classifier	90	0.78	99%	0.68	89%
EMBER	SVC	90	0.83	98%	0.65	89%
ToN-IoT Scanning	GP classifier	90	0.84	100%	0.54	99%
ToN-IoT Scanning	SVC	90	0.75	100%	0.59	99%
UNSW-NB15 DoS	GP classifier	90	0.55	98%	0.51	96%
UNSW-NB15 DoS	SVC	90	0.55	90%	0.42	96%
UNSW-NB15 Recon.	GP classifier	90	0.26	98%	0.57	94%
UNSW-NB15 Recon.	SVC	90	0.19	83%	0.46	99%

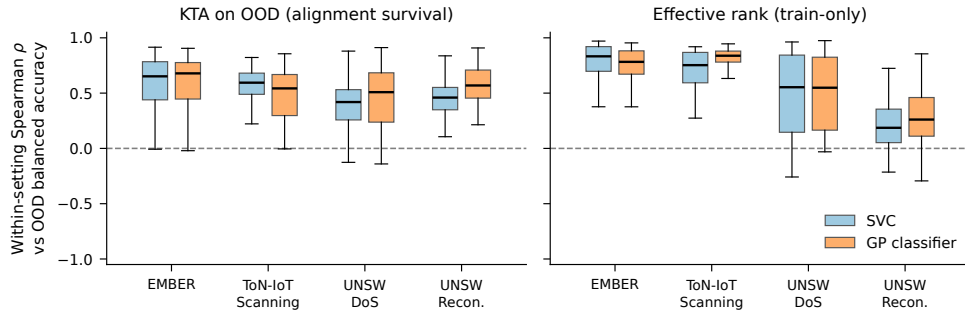


Fig. 3 The mechanism is a cross-dataset regularity. Distribution of within-setting Spearman correlations between each geometry descriptor and OOD balanced accuracy, over all setting-seed units, per dataset and classifier. Alignment survival (left) is positively correlated with OOD accuracy essentially everywhere; effective rank (right) is strong on EMBER and ToN-IoT and weaker, though still predominantly positive, on the UNSW scenarios.

precondition—a kernel whose geometry uses only one or two directions has nothing with which to track label structure into the tails—while alignment survival is the proximate predictor of OOD performance.

A dose-response gradient across both families.

The mechanism makes a prediction that can be tested largely without quantum kernels: kernels with higher effective rank should retain label alignment better under shift.

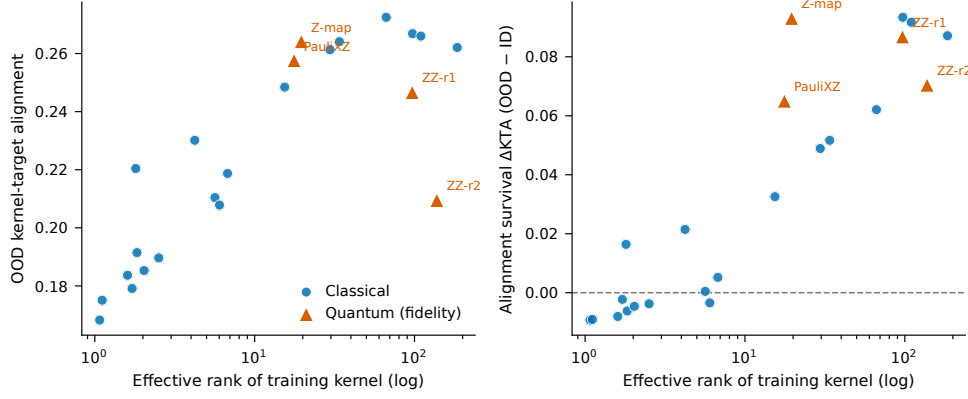


Fig. 4 The cross-family dose-response continuum. Each marker is one kernel—seven base classical kernels, their twelve length-scale variants, and the four fidelity maps—averaged over all EMBER runs and dimensions, positioned by the effective rank of its training Gram matrix. Effective rank orders both OOD kernel-target alignment (left) and its survival from ID to OOD (right) monotonically and identically across families; the fidelity maps (triangles) sit in the middle of the continuum, and short-length-scale classical kernels reach the same high-rank, high-alignment regime. Quantumness is not what distinguishes the robust kernels; effective rank is.

Table 6 confirms a clean monotone gradient across the seven base classical kernels (18 EMBER settings $\times$ 5 dimensions each): as mean effective rank rises from 1.19 (linear) through 2.76 (RBF) and 6–8 (Matérn) to 17.7 (Laplacian), the mean ID-to-OOD change in alignment rises monotonically from -0.017 to $+0.019$, and the fraction of cells where alignment *improves* under shift doubles from 32% to 66%. Adding the twelve length-scale variants and the four fidelity maps extends this into the full continuum of Figure 4. Across all 23 kernels the within-run Spearman correlation between effective rank and OOD alignment has median 0.93 on EMBER (positive in 98% of 450 run–dimension units) and 0.44–0.88 on the network scenarios; crucially, it is just as strong—median 0.96—computed *within the classical family alone*, so the law is not an artefact of a quantum/classical split. The fidelity maps occupy the middle of the continuum: each of the four has between three and nine classical kernels with strictly higher effective rank and better OOD alignment. This explains the honest comparison of Section 4.3 exactly: the classical kernel that closes and then reverses the gap is a Laplacian with a shortened length scale, because shortening the scale is precisely

Table 6 Dose-response gradient in the classical family (18 EMBER settings $\times$ 5 dimensions, q-split seed 42). Higher effective rank of the training Gram matrix is associated with better survival—indeed improvement—of kernel-target alignment from ID to OOD.

Kernel	Eff. rank	Δ KTA (OOD–ID)	Improves
Linear	1.19	−0.017	32%
Polynomial (2)	1.29	−0.017	32%
Polynomial (3)	1.41	−0.015	38%
RBF (scale)	2.76	−0.011	38%
Matérn 5/2	6.41	+0.000	46%
Matérn 3/2	7.71	+0.004	47%
Laplacian	17.69	+0.019	66%

the operation that raises effective rank. The mechanism predicts the winning classical kernel, and it is right.

The mechanism is not an artefact of circularity.

A natural objection is that our proximate predictor—kernel-target alignment on the OOD split—is computed from the same OOD labels used to score accuracy, so their correlation could be mechanical. Three controls rule this out (Table 7 and Figure 5). First, a label-permutation null: recomputing OOD alignment with the OOD labels shuffled (100 permutations per kernel) gives a null centered near 0.005, whereas the observed alignment (0.08–0.45) exceeds it in 100% of kernel cells across every dataset—the alignment is a real property of the label structure, not of chance. Second, cross-fitting: we split each OOD set into halves A and B , measure alignment on A and accuracy on the disjoint half B , and correlate them within each run; the correlation remains positive—median 0.31–0.85 across datasets, positive in 78–100% of units—so alignment measured on independent data still predicts accuracy. Third, and most conservatively, the purely train-side, label-free effective rank predicts OOD accuracy on half B with median Spearman 0.26–0.91: the predictive part of the mechanism needs no OOD labels at all. The weakest scenario, consistent with every other analysis, is UNSW-NB15 Reconnaissance.

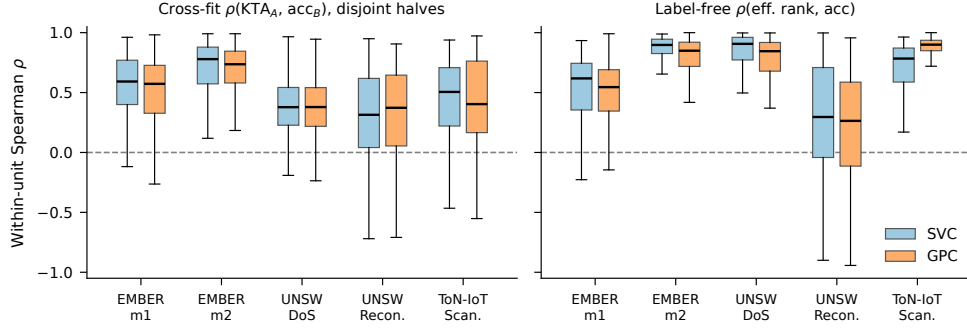


Fig. 5 The mechanism survives circularity controls. Within-unit Spearman correlations, per dataset and classifier. Left: cross-fitted alignment (measured on OOD half A) versus accuracy on the disjoint half B —positive almost everywhere, so the alignment–accuracy link is not an artefact of shared labels. Right: the label-free effective rank of the training kernel versus OOD accuracy, positive throughout. Boxes span the interquartile range; whiskers the 5th–95th percentiles.

Table 7 Anti-circularity controls for the mechanism (per dataset, all kernels, five q-split seeds). “Obs. KTA” is the median observed OOD alignment; “Null” its label-permutation mean; “Sig.” the fraction of kernel cells whose observed alignment exceeds the null at the 5% level. “Cross-fit ρ ” correlates alignment on OOD half A with accuracy on the disjoint half B ; “Rank ρ ” correlates label-free effective rank with OOD accuracy (both averaged over SVC and GPC).

Dataset	Obs. KTA	Null	Sig.	Cross-fit ρ	Rank ρ
EMBER m1 (sparsity)	0.105	0.006	100%	+0.58	+0.58
EMBER m2 (centroid)	0.079	0.006	100%	+0.76	+0.87
UNSW-NB15 DoS	0.288	0.005	100%	+0.38	+0.88
UNSW-NB15 DoS (m2)	0.218	0.005	100%	+0.84	+0.52
UNSW-NB15 Recon.	0.161	0.005	100%	+0.34	+0.28
UNSW-NB15 Recon. (m2)	0.224	0.005	100%	+0.81	+0.46
ToN-IoT Scanning	0.453	0.004	100%	+0.45	+0.84
ToN-IoT Scanning (m2)	0.230	0.005	100%	+0.85	+0.91

What the geometry does not do.

Two negative results complete the mechanistic picture, and both are informative. First, the geometric difference of Huang et al. [Huang et al. \(2021\)](#)—the quantity a reader versed in the quantum-kernel literature would reach for first—does not predict which family wins a setting; at moderate regularization it even correlates *negatively* with the observed OOD delta. This is consistent with its definition: g bounds the potential for a separation between the families, not its sign, so a quantum kernel can differ greatly

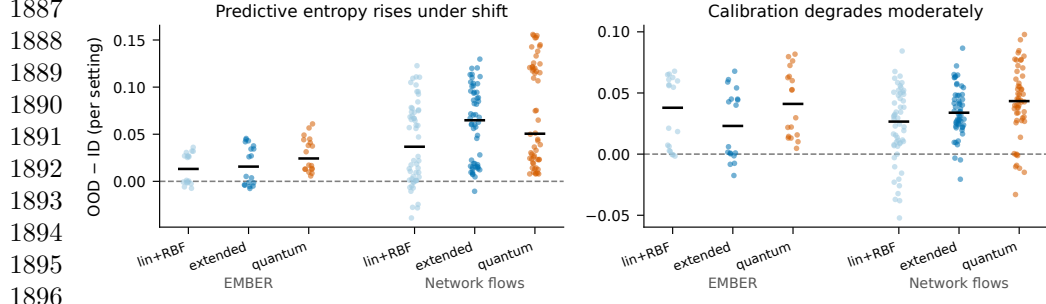


Fig. 6 Predictive uncertainty under shift. Per-setting ID-to-OOD change in mean predictive entropy (left) and expected calibration error (right) for the selected configuration of each family under the GP classifier; black bars mark medians. The qualitative behavior is what one wants from a probabilistic detector—entropy rises under shift for every family—and is shared across families.

from the classical reference and be worse. Second, we found no reliable fully a-priori selection rule: choosing the quantum configuration by training-set alignment alone does not identify the settings where the family will win. The geometry explains and diagnoses; it does not yet provide a deployment-time oracle, and we flag this honestly as an open problem.

4.5 Predictive uncertainty under shift

The GP classifier’s calibrated probabilities add a complementary robustness lens (Figure 6). Averaged over the selected configurations, predictive entropy rises from ID to OOD for every family and dataset group—the qualitative behavior one wants from a probabilistic detector facing drift—and rises most for the quantum kernels (e.g., $+0.029$ versus $+0.014$ – $+0.018$ for the classical families on EMBER; $+0.072$ versus $+0.041$ – $+0.061$ on network flows). Calibration degrades moderately for all families (ECE increases of $+0.02$ – $+0.05$), with the quantum family’s ECE increase slightly larger than the classical families’. In other words, the quantum kernels’ OOD accuracy advantage does not come with hidden overconfidence—their uncertainty grows under shift at least as fast as the classical kernels’—but neither do they calibrate better; the advantage is in the decision geometry, not in the probability calibration.

5 Discussion

5.1 What is, and is not, a quantum advantage here

The study’s central interpretive move is to replace the question “do quantum kernels beat classical kernels under shift?” with “which property of a kernel governs its behavior under shift, and who has it?”. Read through that lens, the results decompose cleanly. The near-uniform advantage over linear and RBF baselines is real, replicable across four datasets and two classifiers, and statistically unambiguous—but it is an advantage of *high-effective-rank*, *alignment-preserving geometry* over the two lowest-rank geometries in the study, a property the fidelity maps possess natively rather than exclusively. Once the classical family is granted the same geometry and the same length-scale tuning freedom, and once selection is not permitted to consult the shifted labels, the out-of-distribution advantage does not merely shrink: it vanishes on EMBER and reverses on network flows, where a short-length-scale Laplacian wins the deployment comparison (Section 4.3). What survives every control we could devise—extended baselines, symmetric length-scale tuning, honest no-label selection, a probabilistic classifier, natural drift, hierarchy-aware inference—is narrower and cleaner: the in-distribution separability advantage, universal on EMBER and unmatched by any classical kernel we tried, and the advantage over the customary weak baselines, which is real but reports the weakness of those baselines as much as the strength of the quantum family. On present evidence, in-distribution separability is the defensible quantum contribution; out-of-distribution robustness is governed by geometry that both families can reach.

From the perspective of the distribution-shift literature, this reinforces a general lesson: conclusions about robustness are only as credible as the protocol used to obtain them [Quiñonero-Candela et al. \(2008\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). Our results add a QML-specific corollary: the composition of the classical reference set, the symmetry of tuning freedom, and whether model

selection is allowed to consult the target labels are not implementation details—each on its own can move the quantum-versus-classical OOD verdict from a large, uniform, highly significant advantage to a null result or a significant reversal (Sections 4.2–4.3). A demonstration that fixes any of them in the quantum family’s favor, reported alone, would mislead.

5.2 The ID–OOD dissociation is informative

The contrast between the two selection views is sharper in this study than in its predecessors, and it now has an interpretation. The quantum ID-separability advantage is universal (18 of 18 EMBER settings against every classical reference, with effects exceeding run-to-run noise in most) and is not replicated by any classical kernel we evaluate, including the Laplacian. The OOD advantage, by contrast, is contingent: it depends on the reference family, the symmetry of tuning, and—decisively—whether selection may consult the shifted labels, and under a fair protocol it is absent or reversed. This dissociation is exactly what the mechanism predicts. In-distribution separability rewards expressive geometry per se; robustness rewards geometry whose label alignment *survives* the shift, which is a joint property of the kernel and the shift mechanism. A kernel family can therefore dominate the first criterion while merely competing on the second—and mistaking one for the other is precisely the error the two-view protocol is designed to prevent [Gulrajani and Lopez-Paz \(2021\)](#). The dissociation also speaks to the “accuracy on the line” literature, which documents a strong empirical coupling between ID and OOD accuracy across models and the settings where it breaks [Miller et al. \(2021\)](#): at the level of kernel families under our shift constructions, the coupling breaks systematically, and the geometry identifies the axis along which it does.

5.3 The mechanism in context

The alignment-survival regularity connects several strands of literature that have so far run in parallel. The bandwidth results extend the findings of Shaydulin and Wild [Shaydulin and Wild \(2022\)](#) and Canatar et al. [Canatar et al. \(2023\)](#) from in-distribution generalization to robustness under shift, and give them a comparative twist: length scale is not merely important for each family in isolation, it must be tuned *symmetrically* for a comparison to be fair, and once it is, the classical family reaches the same robust geometry—so asymmetric tuning protocols do not merely bias quantum-versus-classical comparisons, they manufacture the advantage. The failure of the geometric difference [Huang et al. \(2021\)](#) to predict OOD outcomes sharpens its interpretation rather than contradicting it: g was derived as a bound on the potential for in-distribution separations, and our results show empirically that potential-for-difference and benefit-under-shift are distinct quantities—large g is compatible with large losses. Finally, the dose-response gradient within the classical family (Table 6) grounds the mechanism independently of anything quantum: kernels with heavier spectral tails hold their label alignment into the tails of the data distribution, a statement about kernels generally, for which the quantum maps are simply the most extreme instances in our study.

The high-rank finding must also be reconciled with two spectral results that superficially point the other way. Kübler et al. argue that quantum kernels generalize in-distribution only when the relevant spectral mass is concentrated in few components [Kübler et al. \(2021\)](#), and Thanasilp et al. show that fidelity kernels concentrate exponentially—toward effectively rank-one, uninformative Gram matrices—as qubit counts grow [Thanasilp et al. \(2024\)](#). There is no contradiction, but there is a genuine tension that our results locate precisely. Both results concern the asymptotic or i.i.d. regime; our effective ranks (roughly 10–150 on training sets of 10^3 – 4×10^3 samples

at 4–12 qubits) sit far below the concentration regime, and the studied maps demonstrably retain in-distribution power (the 18-of-18 ID advantage). Within that benign band, the quantity that governs *robustness* is how much of the spectrum survives with its label alignment intact under shift—and there, more rank helps. The bandwidth sweep makes the trade-off operational: angle scaling moves kernels along the rank axis, and the selected scales are those that raise rank without approaching the concentration pathology. Alignment-preserving rank is therefore best understood as bounded from both sides—enough spectral richness to track label structure into the tails, not so much that the kernel concentrates into noise.

We emphasize what the mechanism does not yet deliver. It is explanatory and diagnostic—given the Gram matrices of a setting, it accounts for which kernels will generalize under the shift—but our attempts to convert it into an a-priori selection rule from training data alone failed (Section 4.4). Alignment on OOD inputs requires OOD labels; effective rank is train-only but is a precondition, not a guarantee. Turning alignment survival into a deployable model-selection signal, perhaps via unlabeled OOD data or distributionally robust surrogates, is in our view the most interesting open problem this study raises.

5.4 Implications for security machine learning practice

From a practical perspective, the paper suggests three takeaways for security ML. The first is substantive: for OOD-sensitive kernel selection, prefer kernels whose label-aligned geometry survives shift—high effective rank with alignment that holds into the tails. Concretely, a median-heuristic Laplacian kernel, and especially one with a shortened length scale, is a far stronger robustness baseline than the RBF kernel that pipelines default to; on our data it matches or beats the fidelity maps under honest selection on both modalities, at a fraction of the computational cost. Quantum kernels remain of interest for their in-distribution separability, which no classical kernel

we tried reproduces, but our evidence does not support recommending them over a well-tuned classical kernel when out-of-distribution robustness is the objective.

The second is methodological. Independently of the preferred kernel family, the protocol used here provides a reusable blueprint for robust evaluation in security settings:

1. keep the classifier and preprocessing fixed, and swap only the kernel;
2. define explicit OOD splits tied to interpretable shift mechanisms, including at least one natural drift mechanism where the data permit;
3. give every family the same hyperparameter freedom, bandwidth included;
4. repeat runs, report variability, and test family-level claims formally;
5. and separate model-selection criteria according to the actual objective (robustness vs. separability).

Security ML frequently suffers from comparisons in which model class, tuning effort, and evaluation conditions change simultaneously; the way the verdict of Sections 4.2–4.3 moves from a large advantage to a reversal as the protocol is made fair is a concrete demonstration of how much the conclusion depends on getting this right.

The third concerns diagnostics: the geometry descriptors are cheap to compute from Gram matrices that the pipeline already builds, and the alignment-survival analysis can be run post hoc on any deployed kernel system with a labeled drift sample. Practitioners should nonetheless resist reading these results as a blanket recommendation to replace classical kernels; the responsible recommendation is to test the candidate families under the expected shift regime, using exactly the kind of protocol described here.

5.5 Limitations and threats to validity

Several limitations define the scope of our conclusions.

2163 *Shift operationalization.*

2164

2165 Our OOD splits cover three reproducible mechanisms—within-class sparsity extremes,
2166 train-geometry extremes, and natural regime drift on network traffic—but not the full
2167 space of real-world drift processes. Explicit temporal splits on timestamped malware,
2168 family-based drift, and adversarially induced shift remain outside the study’s scope,
2171 and the alignment-survival regularity should be re-tested under them rather than
2172 assumed.

2173

2174

2175 *Kernel and encoding coverage.*

2176

2177 The classical family now spans seven kernels plus a bandwidth sweep, and both fam-
2178 ilies receive symmetric tuning freedom, but neither candidate set is exhaustive: the
2179 quantum side is restricted to four commonly used fidelity feature maps at low qubit
2180 counts, and trained or task-adapted encodings, projected kernels, and deeper maps are
2181 not evaluated. The mechanism gives a concrete prediction for such extensions—what
2182 should matter is their Gram-matrix geometry, not their provenance—which future
2183 work can falsify. At substantially larger qubit counts, exponential kernel concentra-
2184 tion [Thanasilp et al. \(2024\)](#) would erode the alignment-preserving rank the mechanism
2185 rewards, so our conclusions should not be extrapolated beyond the low-qubit regime
2186 studied here.

2187

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2189 *Selection over candidate sets.*

2190

2191 The honest protocols (P1 deployment, P2 cross-seed), the oracle (P3), and Best-by-ID
2192 are selection procedures over finite candidate sets, applied identically to both families
2193 and evaluated on run means with formal testing. Residual sensitivity to the candidate
2194 space and search granularity nonetheless remains, and fully nested cross-validated
2195 selection would strengthen the analysis further.

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Simulation, not hardware.

All quantum kernels are computed by exact statevector simulation. The study therefore characterizes the geometry of fidelity kernels, not the behavior of noisy devices: finite sampling of kernel entries, hardware noise, and error mitigation would all perturb the Gram matrices [Wang et al. \(2021\)](#), and exponential concentration implies that with polynomially many shots the very spectral richness our mechanism rewards becomes expensive to resolve at larger scales [Thanasilp et al. \(2024\)](#); the effect of either on alignment survival is unknown and worth measuring. We make no claim about hardware feasibility, resource advantage, or quantum speedup—the only rigorously established quantum-classical learning separation concerns a specifically constructed problem, not natural data [Liu et al. \(2021\)](#). At the dimensions studied, the fidelity kernels are also classically simulable [Schreiber et al. \(2023\)](#); [Jerbi et al. \(2024\)](#), which is precisely what makes them available as practical kernel constructions today.

Dataset scope and feature caveats.

Four benchmark scenarios across two modalities is a substantial widening relative to single-dataset studies, but security is broader still: dynamic malware analysis, encrypted traffic, and non-security modalities remain untested. Within the network scenarios, the feature sets inherited from the public benchmarks retain port numbers and protocol indicators, which are legitimate but known shortcut-prone features in NIDS benchmarks [Engelen et al. \(2021\)](#); because every kernel receives identical features, this affects absolute accuracies, not the family comparisons. Finally, the mechanism’s effective-rank component is itself regime-dependent (strong on EMBER and ToN-IoT, weak on UNSW-Reconnaissance), a gradation we report rather than smooth over.

2255 5.6 Conservative interpretation

2256

2257 In light of these limitations and the broader QML benchmarking literature, the
2258 appropriate interpretation of the paper is deliberately conservative. We do not claim
2259 universal superiority, complexity-theoretic quantum advantage, or hardware relevance.
2260 We claim, with formal statistical support, three narrower things: that fidelity-based
2261 quantum kernels beat the customary linear and RBF baselines under shift essentially
2262 wherever we looked; that this advantage is explained by a geometric property—
2263 high-effective-rank, alignment-preserving geometry—that a well-chosen classical kernel
2264 shares in full, so that under symmetric tuning and honest, no-label selection the OOD
2265 advantage over a geometry-matched classical family dissolves on EMBER and reverses
2266 on network flows; and that what remains as a distinct quantum contribution is in-
2267 distribution separability, universal on EMBER and unmatched by any classical kernel
2268 we tried. This regime-aware view is consistent with recent calls for careful bench-
2269 marking in QML [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#) and, we believe,
2270 is scientifically stronger than either an unqualified advantage claim or an unqualified
2271 debunking: it says what the apparent advantage is made of, why it does not survive a
2272 fair comparison, and how to check for the property that does the work.
2273

2284

2285 6 Conclusion

2286

2287 This paper asked why kernel families behave differently under distribution shift,
2288 using the quantum-versus-classical comparison as both motivation and stress test. A
2289 controlled kernel-swap protocol—classifier, preprocessing, splits, selection logic, and
2290 tuning freedom held fixed while only the kernel varies—was instantiated on four secu-
2291 rity benchmarks spanning two modalities and three drift mechanisms, two classifier
2292 families including a probabilistic one, and 72 principal settings with 15 repeated runs
2293 and formal paired testing.
2294

2299

2300

The empirical arc has three movements. Fidelity-based quantum kernels beat the customary linear and RBF baselines under shift nearly everywhere—the result the optimistic literature reports. That advantage, however, does not survive a fair test: once the classical family is enlarged with heavy-tailed kernels granted the same length-scale freedom, and configurations are selected without consulting the shifted labels, the out-of-distribution advantage dissolves on EMBER and reverses on network flows, where a short-length-scale Laplacian wins the deployment comparison. A single mechanism accounts for the whole pattern: out-of-distribution kernel-target alignment is ordered by the effective rank of the training kernel, identically across both families (median within-setting Spearman $\rho \approx 0.93$ on EMBER, positive in 98% of units and just as strong within the classical family alone), and the fidelity maps sit in the middle of that continuum. The mechanism survives label-permutation and cross-fitting controls, explains why a shortened-scale Laplacian is exactly the classical kernel that closes the gap, and makes falsifiable predictions for kernel constructions we did not test.

The contribution is therefore threefold. Empirically, the study provides the broadest controlled, variability-aware regime map of quantum-versus-classical kernel behavior under shift that we are aware of, whose central finding is a negative one obtained under fair conditions—the apparent OOD advantage does not survive symmetric tuning and honest selection—alongside the non-predictiveness of the geometric difference and the absence of an a-priori selection rule. Mechanistically, it converts an observation into an explanation with cross-dataset support that withstands circularity controls. Methodologically, it leaves behind a fully reproducible protocol—and a cautionary demonstration that the quantum-versus-classical OOD verdict moves from a large, uniform advantage to a reversal as the protocol is made fair, which we offer as a concrete standard for future QML benchmarking.

2347 The most important open problems are two. Practically, turning alignment survival
 2348 into a deployment-time selection signal without OOD labels would convert the diag-
 2349 nosis into a tool; the security community’s label-free drift detectors [Yang et al. \(2021\)](#)
 2350 suggest this is not hopeless. Theoretically, the dose-response relationship between spec-
 2351 tral tails and alignment survival calls for a formal account, and the ingredients exist:
 2352 task-model alignment theory explains in-distribution kernel generalization through
 2353 spectrum–target compatibility [Bordelon et al. \(2020\)](#); [Canatar et al. \(2021\)](#), and recent
 2354 minimax analyses characterize RKHS regression under covariate shift with bounded
 2355 density ratios [Ma et al. \(2023\)](#). Extending that line to the survival of aligned spectral
 2356 mass under the shift models studied here—asking under which conditions high effec-
 2357 tive rank provably protects kernel-target alignment—is a concrete research program
 2358 rather than a gap. Answering either question would benefit classical and quantum
 2359 kernel methods alike—which is, in the end, the point: the productive question is not
 2360 whether one family dominates, but which similarity geometry aligns best with the
 2361 shift regime one actually faces.

2372 2373 **Appendix A Additional Results**

2374 This appendix reports the full setting-level *oracle* (Best-by-OOD, protocol P3) com-
 2375 parisons between the selected quantum kernel and the best kernel of the *extended*
 2376 classical family at fixed classical bandwidth, retained for completeness; the honest,
 2377 symmetrically tuned per-scenario deltas that ground our conclusions are in Tables [3](#)
 2378 and [4](#). For both classifiers and all 72 principal settings, every value is a mean $\pm$ stan-
 2379 dard deviation over 15 repeated runs; Effect denotes $\Delta_{\text{OOD}}/\sqrt{\sigma_C^2 + \sigma_Q^2}$. Size labels
 2380 denote $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$: S = (1k, 500, 500), M = (2k, 1k, 1k), L = (4k, 1.8k, 1.8k).
 2381 Shift labels for the network scenarios abbreviate scenario/mechanism: UD = UNSW-
 2382 NB15 DoS, UR = UNSW-NB15 Reconnaissance, TS = ToN-IoT Scanning; /m2 is the
 2383 centroid construction and /nat is natural drift. Quantum kernel names carry the angle
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scale after @ where it differs from 1. All tables are generated directly from the public artifact by `scripts/reporting/make_v2_tables.py`.

Declarations

Availability of data and materials. This study is based on the public EMBER Anderson and Roth (2018), UNSW-NB15 Moustafa and Slay (2015), and ToN-IoT Alsaedi et al. (2020) benchmark datasets. The datasets supporting the conclusions of this article—derived split definitions, shift constructions, experiment manifests, aggregated results, and summary tables—are available in the Zenodo repository, <https://doi.org/10.5281/zenodo.19147649>. All code used to generate the splits, run the classical and quantum kernel experiments, and reproduce every table and figure in this article is part of the same artifact:

- Project name: Controlled Kernel Evaluation under Distribution Shift
- Project home page: https://github.com/roberto-fernandez-barrios/kernel_shift_framework
- Archived version: <https://doi.org/10.5281/zenodo.19147649>
- Operating system(s): platform independent
- Programming language: Python (3.12)
- Other requirements: pinned environment provided (`environment.lock.yml`); qiskit 2.3.1
- License: BSD-3-Clause
- Any restrictions to use by non-academics: none

Competing interests. The authors declare that they have no competing interests.

Funding. The authors received no specific funding for the submitted work.

Authors' contributions. RFB: conceptualization, methodology, software, data curation, formal analysis, visualization, investigation, writing—original draft,

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2441 **Table A1** EMBER, SVC: full Best-by-ODD comparison against the extended classical family.

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Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
m1	42	S	Laplacian (d4)	PauliXZ (d12)	0.756 ± 0.007	0.758 ± 0.020	+0.002	+0.09
m1	42	M	Laplacian (d12)	ZZ-r1 (d12)	0.753 ± 0.019	0.761 ± 0.007	+0.009	+0.42
m1	42	L	Laplacian (d12)	ZZ-r1 (d12)	0.750 ± 0.014	0.757 ± 0.008	+0.006	+0.38
m1	123	S	Laplacian (d12)	PauliXZ (d12)	0.762 ± 0.019	0.750 ± 0.020	-0.012	-0.43
m1	123	M	Laplacian (d12)	ZZ-r1 (d10)	0.760 ± 0.027	0.756 ± 0.029	-0.003	-0.08
m1	123	L	Laplacian (d12)	ZZ-r1 (d10)	0.751 ± 0.019	0.753 ± 0.014	+0.002	+0.09
m1	999	S	Laplacian (d12)	PauliXZ (d12)	0.767 ± 0.028	0.768 ± 0.022	+0.000	+0.00
m1	999	M	Matérn-5/2 (d12)	ZZ-r1 (d12)	0.755 ± 0.015	0.762 ± 0.013	+0.007	+0.34
m1	999	L	RBF (d12)	ZZ-r1 (d10)	0.763 ± 0.024	0.753 ± 0.005	-0.010	-0.42
m2	42	S	Laplacian (d10)	ZZ-r1 (d10)	0.762 ± 0.012	0.763 ± 0.018	+0.000	+0.02
m2	42	M	Laplacian (d12)	ZZ-r2 (d12)	0.767 ± 0.012	0.784 ± 0.009	+0.017	+1.10
m2	42	L	Laplacian (d12)	ZZ-r2 (d12)	0.783 ± 0.008	0.795 ± 0.011	+0.012	+0.85
m2	123	S	Laplacian (d12)	ZZ-r1 (d12)	0.762 ± 0.023	0.771 ± 0.019	+0.009	+0.30
m2	123	M	Laplacian (d12)	ZZ-r1 (d12)	0.775 ± 0.017	0.783 ± 0.013	+0.007	+0.35
m2	123	L	Laplacian (d12)	ZZ-r2 (d12)	0.785 ± 0.011	0.794 ± 0.007	+0.009	+0.71
m2	999	S	Laplacian (d12)	ZZ-r1 (d12)	0.769 ± 0.014	0.775 ± 0.022	+0.006	+0.25
m2	999	M	Laplacian (d10)	ZZ-r2 (d12)	0.776 ± 0.016	0.780 ± 0.020	+0.004	+0.14
m2	999	L	Laplacian (d12)	ZZ-r2 (d12)	0.787 ± 0.013	0.803 ± 0.009	+0.015	+0.97

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2464 **Table A2** EMBER, GPC: full Best-by-ODD comparison against the extended classical family.

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Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
m1	42	S	Laplacian (d12)	PauliXZ (d12)	0.767 ± 0.018	0.760 ± 0.017	-0.007	-0.29
m1	42	M	Laplacian (d12)	ZZ-r1 (d12)	0.765 ± 0.021	0.766 ± 0.011	+0.002	+0.07
m1	42	L	RBF (d10)	PauliXZ (d12)	0.764 ± 0.015	0.761 ± 0.011	-0.003	-0.16
m1	123	S	Laplacian (d12)	PauliXZ (d12)	0.765 ± 0.016	0.753 ± 0.025	-0.012	-0.41
m1	123	M	Laplacian (d12)	PauliXZ (d12)	0.775 ± 0.035	0.772 ± 0.023	-0.004	-0.09
m1	123	L	Matérn-5/2 (d10)	ZZ-r2 (d12)	0.758 ± 0.017	0.754 ± 0.014	-0.004	-0.18
m1	999	S	Laplacian (d12)	PauliXZ (d12)	0.766 ± 0.028	0.762 ± 0.027	-0.004	-0.11
m1	999	M	Matérn-5/2 (d12)	Z-map (d12)	0.768 ± 0.016	0.776 ± 0.009	+0.008	+0.43
m1	999	L	RBF (d12)	ZZ-r1 (d8)	0.763 ± 0.009	0.754 ± 0.008	-0.008	-0.74
m2	42	S	Laplacian (d12)	ZZ-r1 (d12)	0.762 ± 0.024	0.764 ± 0.015	+0.002	+0.06
m2	42	M	Laplacian (d12)	ZZ-r2 (d12)	0.767 ± 0.013	0.778 ± 0.008	+0.010	+0.68
m2	42	L	Laplacian (d12)	ZZ-r2 (d12)	0.783 ± 0.008	0.793 ± 0.009	+0.010	+0.85
m2	123	S	Laplacian (d12)	ZZ-r2 (d12)	0.767 ± 0.025	0.772 ± 0.015	+0.005	+0.19
m2	123	M	Laplacian (d12)	ZZ-r1 (d12)	0.772 ± 0.020	0.783 ± 0.015	+0.011	+0.44
m2	123	L	Laplacian (d12)	ZZ-r2 (d12)	0.780 ± 0.010	0.791 ± 0.010	+0.012	+0.83
m2	999	S	Laplacian (d12)	ZZ-r1 (d10)	0.767 ± 0.021	0.775 ± 0.021	+0.008	+0.27
m2	999	M	Laplacian (d12)	ZZ-r2 (d12)	0.776 ± 0.016	0.780 ± 0.025	+0.004	+0.13
m2	999	L	Laplacian (d12)	ZZ-r2 (d12)	0.786 ± 0.013	0.798 ± 0.011	+0.012	+0.70

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Table A3 Network flows, SVC: full Best-by-OOD comparison against the extended classical family.

Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
TS/m2	42	S	Laplacian (d12)	ZZ-r1 (d6)	0.941 ± 0.013	0.954 ± 0.009	+0.014	+0.90
TS/m2	42	M	Laplacian (d12)	ZZ-r1 (d10)	0.941 ± 0.014	0.958 ± 0.005	+0.017	+1.15
TS/m2	42	L	Laplacian (d12)	ZZ-r1 (d8)	0.959 ± 0.009	0.964 ± 0.006	+0.005	+0.44
TS/m2	123	S	Laplacian (d12)	ZZ-r1 (d8)	0.943 ± 0.017	0.959 ± 0.011	+0.015	+0.75
TS/m2	123	M	Laplacian (d12)	ZZ-r1 (d6)	0.954 ± 0.007	0.958 ± 0.006	+0.003	+0.35
TS/m2	123	L	Laplacian (d10)	ZZ-r1 (d8)	0.955 ± 0.003	0.959 ± 0.003	+0.003	+0.72
TS/m2	999	S	Laplacian (d10)	ZZ-r1 (d10)	0.936 ± 0.011	0.957 ± 0.003	+0.021	+1.80
TS/m2	999	M	Laplacian (d12)	ZZ-r1 (d8)	0.954 ± 0.009	0.962 ± 0.007	+0.008	+0.72
TS/m2	999	L	Laplacian (d12)	ZZ-r1 (d10)	0.965 ± 0.004	0.969 ± 0.003	+0.003	+0.71
TS/nat	42	S	Laplacian (d8)	ZZ-r1 (d10)	0.987 ± 0.003	0.988 ± 0.005	+0.001	+0.10
TS/nat	42	M	Laplacian (d10)	ZZ-r1 (d12)	0.989 ± 0.004	0.990 ± 0.002	+0.001	+0.24
TS/nat	42	L	Laplacian (d12)	Z-map (d12)	0.989 ± 0.005	0.991 ± 0.003	+0.002	+0.31
TS/nat	123	S	Laplacian (d8)	ZZ-r1 (d12)	0.990 ± 0.005	0.992 ± 0.005	+0.002	+0.36
TS/nat	123	M	Laplacian (d12)	Z-map (d10)	0.989 ± 0.003	0.992 ± 0.002	+0.003	+0.61
TS/nat	123	L	Laplacian (d10)	Z-map (d12)	0.990 ± 0.002	0.991 ± 0.003	+0.002	+0.40
TS/nat	999	S	Laplacian (d12)	Z-map (d12)	0.979 ± 0.006	0.985 ± 0.002	+0.006	+1.05
TS/nat	999	M	Laplacian (d10)	Z-map (d10)	0.990 ± 0.002	0.991 ± 0.004	+0.002	+0.38
TS/nat	999	L	Laplacian (d10)	Z-map (d12)	0.992 ± 0.002	0.992 ± 0.003	+0.000	+0.12
UD/m2	42	S	Laplacian (d12)	PauliXZ (d12)	0.920 ± 0.016	0.919 ± 0.014	-0.001	-0.06
UD/m2	42	M	Laplacian (d12)	PauliXZ (d12)	0.934 ± 0.017	0.937 ± 0.007	+0.003	+0.16
UD/m2	42	L	Laplacian (d12)	PauliXZ (d10)	0.942 ± 0.008	0.940 ± 0.006	-0.002	-0.18
UD/m2	123	S	Laplacian (d12)	PauliXZ (d12)	0.917 ± 0.020	0.926 ± 0.013	+0.008	+0.33
UD/m2	123	M	Laplacian (d12)	PauliXZ (d12)	0.943 ± 0.007	0.940 ± 0.010	-0.004	-0.28
UD/m2	123	L	Laplacian (d12)	Z-map (d12)	0.948 ± 0.003	0.947 ± 0.005	-0.001	-0.20
UD/m2	999	S	Laplacian (d12)	PauliXZ (d12)	0.923 ± 0.023	0.927 ± 0.009	+0.004	+0.16
UD/m2	999	M	Laplacian (d12)	PauliXZ (d12)	0.935 ± 0.013	0.939 ± 0.007	+0.004	+0.26
UD/m2	999	L	Laplacian (d12)	Z-map (d12)	0.947 ± 0.004	0.946 ± 0.004	-0.001	-0.22
UD/nat	42	S	Linear (d4)	ZZ-r2 (d10)	0.781 ± 0.040	0.809 ± 0.017	+0.028	+0.65
UD/nat	42	M	Laplacian (d12)	Z-map (d12)	0.829 ± 0.019	0.838 ± 0.014	+0.009	+0.38
UD/nat	42	L	Laplacian (d12)	Z-map (d12)	0.829 ± 0.011	0.829 ± 0.008	+0.000	+0.01
UD/nat	123	S	Laplacian (d12)	ZZ-r2 (d6)	0.775 ± 0.026	0.818 ± 0.018	+0.043	+1.37
UD/nat	123	M	Laplacian (d12)	ZZ-r1 (d10)	0.790 ± 0.020	0.812 ± 0.007	+0.022	+1.04
UD/nat	123	L	Laplacian (d12)	ZZ-r1 (d12)	0.809 ± 0.012	0.825 ± 0.011	+0.016	+0.97
UD/nat	999	S	Linear (d4)	ZZ-r2 (d6)	0.759 ± 0.051	0.800 ± 0.018	+0.042	+0.77
UD/nat	999	M	Laplacian (d12)	ZZ-r2 (d10)	0.788 ± 0.016	0.809 ± 0.014	+0.021	+0.95
UD/nat	999	L	Laplacian (d12)	ZZ-r1 (d12)	0.804 ± 0.006	0.826 ± 0.013	+0.023	+1.63
UR/m2	42	S	Laplacian (d12)	Z-map (d12)	0.909 ± 0.025	0.909 ± 0.015	-0.000	-0.01
UR/m2	42	M	Laplacian (d12)	PauliXZ (d12)	0.931 ± 0.019	0.935 ± 0.008	+0.004	+0.18
UR/m2	42	L	Laplacian (d12)	PauliXZ (d12)	0.944 ± 0.005	0.943 ± 0.006	-0.001	-0.12
UR/m2	123	S	Laplacian (d12)	PauliXZ (d12)	0.946 ± 0.014	0.942 ± 0.007	-0.004	-0.24
UR/m2	123	M	Laplacian (d12)	PauliXZ (d12)	0.951 ± 0.011	0.948 ± 0.008	-0.004	-0.26
UR/m2	123	L	Laplacian (d12)	Z-map (d12)	0.954 ± 0.007	0.954 ± 0.006	-0.000	-0.03
UR/m2	999	S	Laplacian (d12)	Z-map (d12)	0.915 ± 0.016	0.918 ± 0.014	+0.003	+0.12
UR/m2	999	M	Laplacian (d12)	Z-map (d12)	0.947 ± 0.006	0.944 ± 0.008	-0.003	-0.33
UR/m2	999	L	Laplacian (d12)	Z-map (d12)	0.956 ± 0.003	0.948 ± 0.007	-0.008	-1.03
UR/nat	42	S	Linear (d12)	ZZ-r2 (d6)	0.770 ± 0.019	0.778 ± 0.013	+0.008	+0.36
UR/nat	42	M	Laplacian (d10)	ZZ-r1 (d4)	0.809 ± 0.035	0.815 ± 0.010	+0.007	+0.18
UR/nat	42	L	Laplacian (d10)	ZZ-r1 (d4)	0.812 ± 0.020	0.815 ± 0.008	+0.003	+0.12
UR/nat	123	S	Linear (d12)	ZZ-r1 (d6)	0.795 ± 0.012	0.797 ± 0.018	+0.002	+0.11
UR/nat	123	M	Linear (d10)	ZZ-r1 (d4)	0.800 ± 0.018	0.802 ± 0.024	+0.003	+0.09
UR/nat	123	L	Laplacian (d12)	ZZ-r1 (d4)	0.818 ± 0.018	0.811 ± 0.009	-0.007	-0.34
UR/nat	999	S	Linear (d12)	ZZ-r1 (d6)	0.780 ± 0.020	0.781 ± 0.008	+0.000	+0.02
UR/nat	999	M	Linear (d12)	ZZ-r2 (d6)	0.783 ± 0.014	0.809 ± 0.015	+0.026	+1.26
UR/nat	999	L	Laplacian (d8)	ZZ-r2 (d6)	0.804 ± 0.015	0.814 ± 0.008	+0.010	+0.59

Table A4 Network flows, GPC: full Best-by-OOD comparison against the extended classical family.

Shift	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Eff.
TS/m2	42	S	Laplacian (d12)	ZZ-r1 (d8)	0.906 ± 0.019	0.954 ± 0.011	$+0.049$	$+2.27$
TS/m2	42	M	Laplacian (d8)	ZZ-r1 (d12)	0.921 ± 0.004	0.957 ± 0.005	$+0.037$	$+5.70$
TS/m2	42	L	Laplacian (d8)	ZZ-r1 (d8)	0.931 ± 0.007	0.964 ± 0.006	$+0.034$	$+3.58$
TS/m2	123	S	Laplacian (d8)	ZZ-r1 (d10)	0.910 ± 0.056	0.958 ± 0.006	$+0.048$	$+0.86$
TS/m2	123	M	Laplacian (d4)	ZZ-r1 (d12)	0.932 ± 0.007	0.958 ± 0.007	$+0.025$	$+2.59$
TS/m2	123	L	Laplacian (d4)	ZZ-r1 (d10)	0.934 ± 0.004	0.960 ± 0.003	$+0.026$	$+4.84$
TS/m2	999	S	Laplacian (d6)	ZZ-r1 (d10)	0.918 ± 0.021	0.954 ± 0.009	$+0.036$	$+1.58$
TS/m2	999	M	Laplacian (d4)	ZZ-r1 (d10)	0.931 ± 0.002	0.961 ± 0.006	$+0.030$	$+4.49$
TS/m2	999	L	Laplacian (d8)	ZZ-r1 (d10)	0.941 ± 0.003	0.968 ± 0.003	$+0.027$	$+6.65$
TS/nat	42	S	Laplacian (d8)	ZZ-r1 (d8)	0.981 ± 0.005	0.990 ± 0.002	$+0.009$	$+1.67$
TS/nat	42	M	Laplacian (d8)	ZZ-r1 (d10)	0.987 ± 0.002	0.991 ± 0.003	$+0.004$	$+1.08$
TS/nat	42	L	Laplacian (d8)	ZZ-r1 (d12)	0.983 ± 0.003	0.992 ± 0.002	$+0.009$	$+2.30$
TS/nat	123	S	Laplacian (d8)	ZZ-r1 (d8)	0.981 ± 0.017	0.992 ± 0.003	$+0.011$	$+0.65$
TS/nat	123	M	Laplacian (d8)	ZZ-r1 (d10)	0.985 ± 0.004	0.993 ± 0.002	$+0.008$	$+1.72$
TS/nat	123	L	Laplacian (d8)	ZZ-r1 (d12)	0.985 ± 0.004	0.992 ± 0.002	$+0.007$	$+1.64$
TS/nat	999	S	Laplacian (d10)	ZZ-r1 (d10)	0.965 ± 0.027	0.980 ± 0.015	$+0.015$	$+0.48$
TS/nat	999	M	Laplacian (d8)	ZZ-r1 (d10)	0.984 ± 0.005	0.991 ± 0.003	$+0.007$	$+1.20$
TS/nat	999	L	Laplacian (d8)	ZZ-r1 (d10)	0.988 ± 0.002	0.992 ± 0.003	$+0.004$	$+1.41$
UD/m2	42	S	Laplacian (d12)	PauliXZ (d12)	0.894 ± 0.028	0.904 ± 0.016	$+0.009$	$+0.29$
UD/m2	42	M	Laplacian (d12)	PauliXZ (d12)	0.911 ± 0.007	0.934 ± 0.009	$+0.023$	$+1.96$
UD/m2	42	L	Laplacian (d12)	Z-map (d12)	0.927 ± 0.008	0.939 ± 0.010	$+0.013$	$+1.02$
UD/m2	123	S	Laplacian (d12)	Z-map (d12)	0.889 ± 0.019	0.917 ± 0.019	$+0.028$	$+1.04$
UD/m2	123	M	Laplacian (d12)	PauliXZ (d12)	0.916 ± 0.016	0.937 ± 0.007	$+0.022$	$+1.24$
UD/m2	123	L	Laplacian (d12)	Z-map (d12)	0.932 ± 0.014	0.945 ± 0.007	$+0.013$	$+0.86$
UD/m2	999	S	Laplacian (d12)	Z-map (d12)	0.878 ± 0.022	0.911 ± 0.031	$+0.033$	$+0.87$
UD/m2	999	M	Laplacian (d12)	PauliXZ (d12)	0.914 ± 0.011	0.931 ± 0.009	$+0.017$	$+1.19$
UD/m2	999	L	Laplacian (d12)	PauliXZ (d12)	0.936 ± 0.009	0.944 ± 0.003	$+0.008$	$+0.82$
UD/nat	42	S	Laplacian (d12)	ZZ-r2 (d6)	0.764 ± 0.023	0.801 ± 0.023	$+0.037$	$+1.13$
UD/nat	42	M	Laplacian (d12)	PauliXZ (d12)	0.813 ± 0.010	0.829 ± 0.008	$+0.016$	$+1.20$
UD/nat	42	L	Laplacian (d12)	Z-map (d12)	0.818 ± 0.010	0.831 ± 0.009	$+0.013$	$+0.97$
UD/nat	123	S	Laplacian (d12)	ZZ-r2 (d10)	0.775 ± 0.027	0.813 ± 0.017	$+0.038$	$+1.20$
UD/nat	123	M	Laplacian (d12)	ZZ-r2 (d6)	0.781 ± 0.020	0.805 ± 0.016	$+0.024$	$+0.93$
UD/nat	123	L	Laplacian (d12)	ZZ-r1 (d12)	0.800 ± 0.005	0.816 ± 0.006	$+0.016$	$+2.09$
UD/nat	999	S	Linear (d4)	ZZ-r2 (d6)	0.766 ± 0.053	0.797 ± 0.021	$+0.032$	$+0.55$
UD/nat	999	M	Laplacian (d12)	ZZ-r2 (d12)	0.784 ± 0.018	0.806 ± 0.013	$+0.022$	$+1.02$
UD/nat	999	L	Laplacian (d12)	ZZ-r2 (d12)	0.805 ± 0.006	0.822 ± 0.012	$+0.017$	$+1.31$
UR/m2	42	S	Laplacian (d12)	Z-map (d12)	0.880 ± 0.021	0.902 ± 0.022	$+0.022$	$+0.73$
UR/m2	42	M	Laplacian (d12)	Z-map (d12)	0.901 ± 0.008	0.925 ± 0.009	$+0.024$	$+1.99$
UR/m2	42	L	Laplacian (d12)	PauliXZ (d12)	0.929 ± 0.012	0.942 ± 0.003	$+0.013$	$+1.05$
UR/m2	123	S	Laplacian (d12)	Z-map (d12)	0.905 ± 0.012	0.926 ± 0.023	$+0.021$	$+0.81$
UR/m2	123	M	Laplacian (d12)	Z-map (d12)	0.914 ± 0.010	0.946 ± 0.010	$+0.032$	$+2.29$
UR/m2	123	L	Laplacian (d12)	PauliXZ (d12)	0.946 ± 0.004	0.953 ± 0.006	$+0.007$	$+0.91$
UR/m2	999	S	Laplacian (d10)	PauliXZ (d12)	0.883 ± 0.023	0.910 ± 0.025	$+0.027$	$+0.81$
UR/m2	999	M	Laplacian (d12)	Z-map (d12)	0.912 ± 0.004	0.941 ± 0.008	$+0.029$	$+3.13$
UR/m2	999	L	Laplacian (d10)	PauliXZ (d12)	0.937 ± 0.014	0.949 ± 0.003	$+0.012$	$+0.84$
UR/nat	42	S	Poly-2 (d4)	PauliXZ (d6)	0.760 ± 0.020	0.784 ± 0.024	$+0.025$	$+0.79$
UR/nat	42	M	Laplacian (d10)	ZZ-r1 (d4)	0.795 ± 0.040	0.817 ± 0.013	$+0.022$	$+0.51$
UR/nat	42	L	Laplacian (d8)	ZZ-r1 (d4)	0.796 ± 0.007	0.813 ± 0.005	$+0.017$	$+1.95$
UR/nat	123	S	Poly-2 (d12)	ZZ-r2 (d6)	0.780 ± 0.024	0.797 ± 0.016	$+0.017$	$+0.59$
UR/nat	123	M	Linear (d12)	ZZ-r1 (d4)	0.794 ± 0.014	0.803 ± 0.020	$+0.008$	$+0.34$
UR/nat	123	L	Laplacian (d12)	ZZ-r2 (d12)	0.810 ± 0.023	0.812 ± 0.019	$+0.002$	$+0.06$
UR/nat	999	S	Poly-2 (d12)	ZZ-r2 (d6)	0.777 ± 0.016	0.782 ± 0.005	$+0.005$	$+0.30$
UR/nat	999	M	Poly-2 (d12)	ZZ-r2 (d6)	0.780 ± 0.017	0.814 ± 0.016	$+0.034$	$+1.51$
UR/nat	999	L	Laplacian (d8)	ZZ-r1 (d4)	0.790 ± 0.010	0.807 ± 0.011	$+0.017$	$+1.10$

writing—review and editing. IPL: supervision, validation, writing—review and editing. 2577
 AGS: validation, project administration, writing—review and editing. PGB: supervi- 2578
 sion, resources, writing—review and editing. All authors read and approved the final 2579
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Acknowledgements. Not applicable. 2583
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